Gelfand seminar — chapter 1

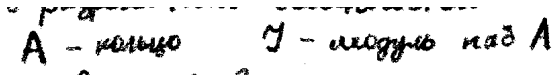
Opening of the reader. Source frames 1–15. Text and formulas are typed; original details accompany every uncertain reading. Working excerpt; the complete chapter is still in preparation.

GELFAND-P000001

Monday, 21/IX. Cartesian and Boolean products of groups.

“Hironaka: on resolution of singularities.” Gelfand

A is a ring, $\mathcal{J}$ is a module over A.

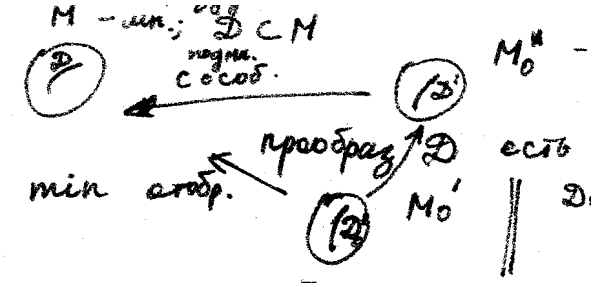


Original source details: GELFAND-P000001

Uncertain reading. The letter for the module and ideal is rendered as $\mathcal{J}$; $\mathcal{I}$ or J are possible.

Blow-up — ? M is a manifold; $D \subset M$. [Beneath D:] a subset with singularities.

M_0^H is the blow-up. Blowing-up. The inverse image of D is D', a manifold of dimension $n-1$. D_0 is $(n-1)$-dimensional.



Original source details: GELFAND-P000001

Diagram and typed description. Three enclosed drawings: at left a region with D; at upper right a region with D' and the label M_0^H; at lower right a region and the label M_0'. The upper arrow points left; the lower oblique arrow also points left, with the label “continuous map” [source abbreviation: непр. отобр.]. The precise inner curves, labels and the connection of the lower region to the upper one require comparison with the scan.



Original source details: GELFAND-P000001

Uncertain reading. The upper index on M_0 resembles H; the lower region’s label resembles M_0'.

The construction is Grothendieck’s. A is a ring, $\mathcal{J}$ is a module. Example. A is a ring of functions; $\mathcal{J}$ is the ideal of functions taking the value 0 on D.

One obtains a sheaf of rings and a sheaf of modules. A is a ring, $\mathcal{J}$ is an ideal. Construct X, O_X a sheaf of rings, $\mathcal{J}_X$ a sheaf of ideals, so that

$$(A, \mathcal{J}) \longleftarrow (X, O_X, \mathcal{J}_X).$$

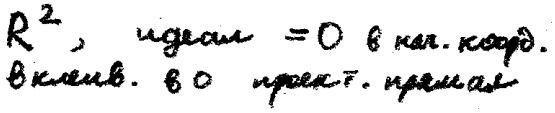
[Beneath the arrow:] linear map.

$(A, \mathcal{J})$ is a sheaf generated by 1 ideal.

$\mathcal{J}_X$ is invertible, i.e. an ideal of the ring: $\exists f \in \mathcal{J}_X$ such that $O_X f = \mathcal{J}_X$. [Above the formula, an arrow to $\exists$:] generator.

(The ideal is principal if $\mathcal{J}$ is an ideal.) Then the zero variety is 1 equation.

A is a ring, $\mathcal{J}$ is an ideal. Typical example: R^2, the ideal $= 0$ at the origin.



Original source details: GELFAND-P000001

Uncertain reading. The next line: “Gluing a projective line into 0” [source: Вклеив. в 0 проект. прямой]; the beginning may read “Оклеив.”.

A, $\mathcal{J}$ are modules.

$$\mathcal{J} \otimes_A \mathcal{J}, \qquad \mathcal{J} \otimes_A \mathcal{J} \otimes_A \mathcal{J}, \dots$$

[Above the groups of tensor products:] symmetric; symmetric.

Consider formal sums

$$\sum_{k=0}^{N} a_k z^k.$$

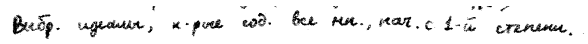
Add and multiply. $x \otimes y = y \otimes x$.

$$a_0 \in A, \qquad a_1 \in \mathcal{J}, \qquad a_2 \in \mathcal{J} \otimes_A \mathcal{J} \quad —$$

GELFAND-P000002

The ring of these sums is $S_A(\mathcal{J})$. Consider homogeneous ideals [inserted above the line:] (graded). (Together with a polynomial it contains all its homogeneous components.)

Consider all homogeneous prime ideals in $S_A(\mathcal{J})$. Выбр. [source abbreviation] ideals which contain all polynomials, starting with degree 1.



Original source details: GELFAND-P000002

Uncertain reading. The end of the line after “contain all” is better read as “polynomials, starting with degree 1”; the middle strokes of the abbreviation “нач.” remain indistinct.

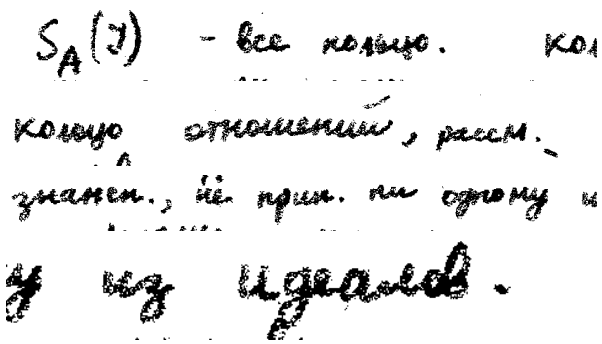
The collection of ideals is X. (The Zariski topology.) O_X, $\mathcal{J}_X$. If U is an open set, then O_U, $\mathcal{J}_U$.



Original source details: GELFAND-P000002

Uncertain reading. After “then”, the first letter has the same form as O_X above, the second as the ideal letter: O_U, $\mathcal{J}_U$; the latter may be $\mathcal{I}_U$.

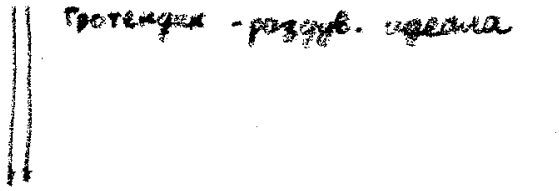
$S_A(\mathcal{J})$ is the whole ring. The ring of fractions, numerators, denominators, not belonging to any of the ideals. Localize the ring at this set. Fractions of degree 1: $\mathcal{J}_U$.



Original source details: GELFAND-P000002

Uncertain reading. In the line about fractions, the abbreviations “numerator–denominator” and whether “not belonging” applies only to the denominator are unclear. No words are supplied from the meaning.

[At right, two downward arrows:] Grothendieck — blow up ideally.



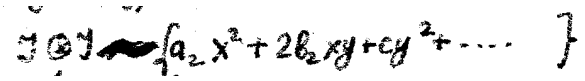
Original source details: GELFAND-P000002

Uncertain reading. The right-hand note reads “Grothendieck — blow up ideally”; the last word may be “идеально” [ideally] or “идеалы” [ideals].

A is the ring of polynomials in 2 variables. $\mathcal{J}$ is the ideal of polynomials with $P(0,0)=0$.

$$A=\{P(x,y)\},\qquad \mathcal{J}=P(x,y)=a_1x+b_1y+\cdots.$$

$$\mathcal{J}\otimes_A\mathcal{J}\ \longleftrightarrow\ \{a_2x^2+2b_2xy+cy^2+\cdots\}.$$



Original source details: GELFAND-P000002

Uncertain reading. The coefficients in the quadratic polynomial read a_2, $2b_2$, c; the index on c and the shape of the thick two-headed arrow are uncertain.

Because of the interchange, one obtains the same as the square of the ideal. Then — the ideal of polynomials equal to 0 at this point. What will it blow up into? ?

$$\sum P_k(x,y)z^k,\qquad P_k(x,y)=ax^k+\cdots.$$

[At right:] Graded Σ of tensor products.

$$A=\{P_0(x,y)\}\longrightarrow S_A(\mathcal{J}),\qquad P_0(x,y)\sim P_0(x,y).$$

The set of prime ideals obtained from the ideal $P(0,0)=0$.

$$S_A\sim\sum P_kz^k,\qquad P_k\in\mathcal{J}^k,\qquad K\sim\sum P_kz^k.$$

It suffices to find the max. ideals of the quotient ring — (to find the ideals containing the given one).

$$S_A/K\ \longleftrightarrow\ a_0+(a_1x-b_1y)z+\cdots.$$



Original source details: GELFAND-P000002

Uncertain reading. In the last formula a minus is visible before b_1y, although the next scan has a plus. The visible minus is retained; the direction of the thick arrow is also uncertain.

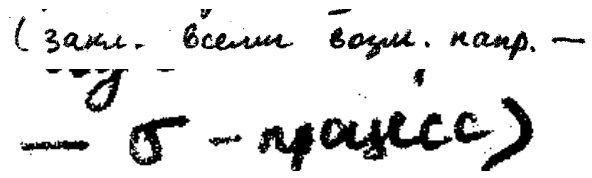
GELFAND-P000003

All homogeneous max. ideals in it will be the points of the extension. There is no constant term (then also 1).

$$(a_1x+b_1y)z.$$

Only 1 linear combination of x and y. (Not 2, since then all.)

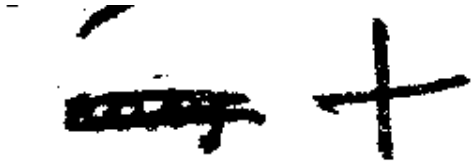
The ideals determine directions in the plane (замк. [source abbreviation] with all possible directions — the σ-process).



Original source details: GELFAND-P000003

Uncertain reading. The abbreviations in parentheses read “замк. всеми возм. напр.”; neither the first word nor the meaning of the sign σ is asserted.

Problem. In the space [cancelled:]



Uncertain reading. The cancelled word before the drawing is illegible.

Original source details: GELFAND-P000003



Original diagram



Typeset recreation

What results from gluing it in?

$$z = 0, \qquad xy = 0, \qquad \mathcal{J} = (z, xy).$$

[A line is drawn beneath the system.]

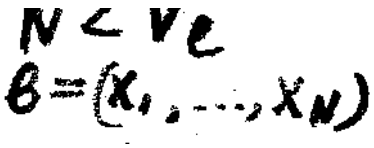
Dobrushin. [Cancelled before “Find”:]



Uncertain reading. The cancelled beginning of the line cannot be read confidently.

Original source details: GELFAND-P000003

Find an example of a gas in which phase transitions occur. A cube of the integer lattice. ℓ is the length. $V_\ell = \ell^\nu$. N points, $N < V_\ell$. They can occupy positions on the lattice $\theta = (x_1, \ldots, x_N)$.



Uncertain reading. The configuration sign is rendered as θ; the handwritten sign also resembles b or Cyrillic “в”.

Original source details: GELFAND-P000003

The Ising model. [Before the formula, a densely cancelled sign:]

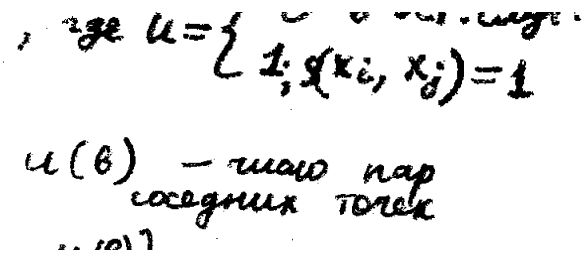


Uncertain reading. The separate cancelled sign before $U(\theta)$ is illegible.

Original source details: GELFAND-P000003

$$U(\theta) = -\frac{1}{2}\sum_{i \neq j} u(x_i, x_j), \qquad u = \begin{cases} 0 & \text{in the other cases,} \\ 1; & \rho(x_i, x_j) = 1. \end{cases}$$

$u(\theta)$ is the number of pairs of neighboring points.



Original source details: GELFAND-P000003

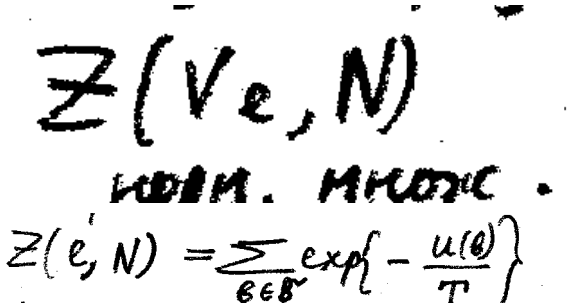
Uncertain reading. In the condition for the second branch a sign resembling ρ is visible; the semicolon after 1 is retained. In the caption below, the first letter resembles lowercase u rather than U.

The basic axiom is the Gibbs distribution.

$$\mathrm{P}(B) = \frac{\sum\limits_{\theta \in B} \exp\{-U(\theta)/T\}}{Z(V_\ell, N)}, \qquad T > 0 \text{ — temperature.}$$

[Beneath the denominator:] normalizing factor.

$$Z(\ell, N) = \sum_{\theta \in \mathcal{B}} \exp\{-U(\theta)/T\}.$$



Original source details: GELFAND-P000003

Uncertain reading. The denominator in the first formula reads $Z(V_\ell, N)$, the next line $Z(\ell, N)$; the set sign beneath the sum resembles $\mathcal{B}$. The notation is not adjusted for consistency.

The molecules attract one another. ([Two filled signs with arrows:] neighbors $>$ their probability.)



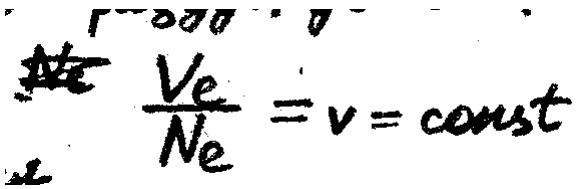
Original source details: GELFAND-P000003

Uncertain reading. In parentheses after “The molecules attract”, there are filled signs and arrows; the reading “neighbors $>$ their probability” is uncertain.

$B \subset \mathcal{B}$ is a collection of configurations.

$$p(\theta) = \frac{\exp\{-U(\theta)/T\}}{Z(\ell, N)}.$$

Take the limit $\ell \to \infty$; the cube expands to ∞. N_ℓ depends on ℓ. [A cancelled sign before the fraction:]



Original source details: GELFAND-P000003

Uncertain reading. The cancelled sign before V_ℓ/N_ℓ is illegible.

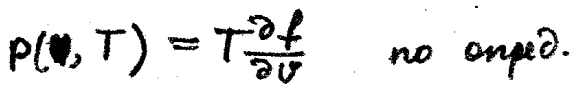
$$\frac{V_\ell}{N_\ell} = v = \text{const}.$$

The lattice grows. One can show that

$$\frac{1}{N_\ell} \log Z(\ell, N_\ell) \longrightarrow f(v, T).$$

GELFAND-P000004

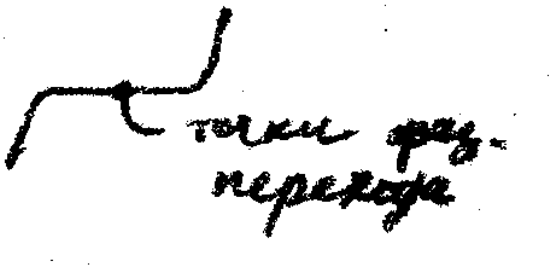
$$p(v,T) = T\frac{\partial f}{\partial v} \qquad \text{by definition.}$$



Original source details: GELFAND-P000004

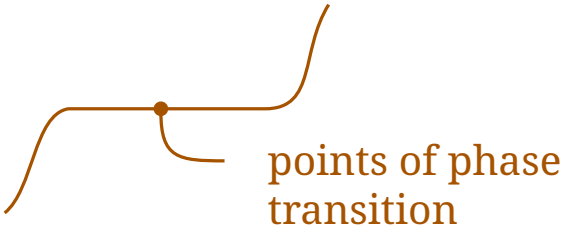
Uncertain reading. The heavily written first argument of p resembles v; no minus is visible before the derivative, and none is supplied.

Phase transition. As for f: it is a monotone, convex function of v. Therefore $p > 0$, and p is a monotone function of v. Phase transition: $p(v, T)$ has a flat segment.



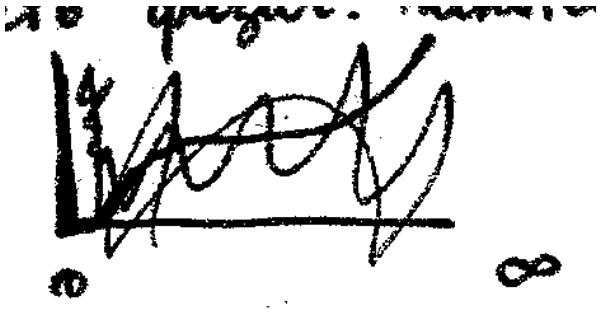
Original source details: GELFAND-P000004

Diagram and typed description. In the upper right-hand field is a bent, S-like curve with an almost horizontal middle segment and an arrow toward it; the caption reads “points of phase transition” [source abbreviation: точки фаз. перехода]. Precise axes and points in the original drawing are not identified confidently.



Typeset sketch; limitations are recorded in the note.

There is a physical hypothesis: 1 transition segment.

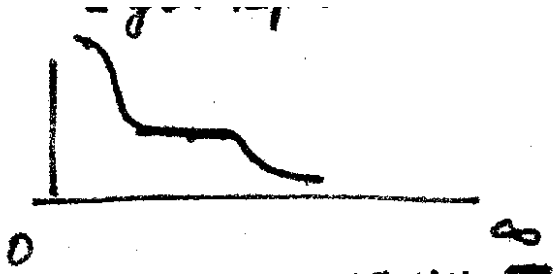


Original source details: GELFAND-P000004

Diagram and typed description. At left is a graph with vertical and horizontal axes, 0 and ∞ by the horizontal axis. Several curves and straight lines are heavily crossed out; the precise cancelled curve and the vertical label need comparison.

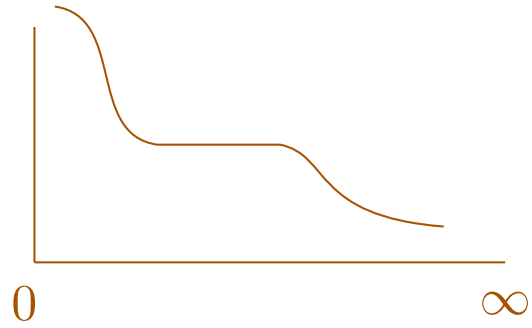


Partial recreation: secure axes only; cancelled curves remain unresolved.



Original source details: GELFAND-P000004

Diagram and typed description. Neighboring drawing at right: a clean descending curve with a flat segment. This is a separate graph, not a reconstruction of the cancelled drawing at left.

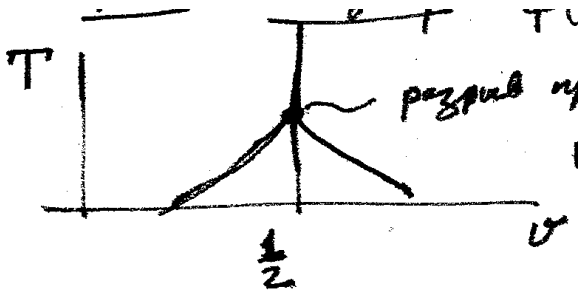


Existing typeset recreation.

A flat segment occurs for sufficiently small T ($< T_{\text{crit}}$).

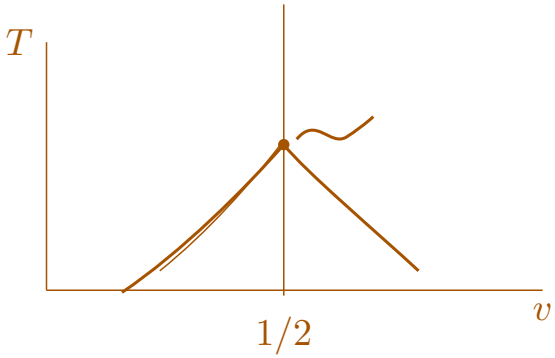
It admits an explicit solution for $n = 1$. Z and $f(v, T)$ are calculated. There are no phase transitions, which is considered physically natural.

$n = 2$. Onsager. $f(v, T)$ is calculated for particular v and T. A discontinuity of the derivative corresponds to phase transitions.



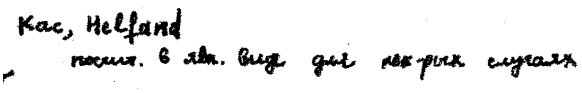
Original source details: GELFAND-P000004

Diagram and typed description. A graph with the T axis upward, the v axis to the right, and the label $1/2$ beneath a vertical through the vertex. The curve rises from the left to the vertex and descends to the right; the vertical line passes through the vertex, with a short wavy segment to its right. The precise geometry and the meaning of the wavy branch have not been established.



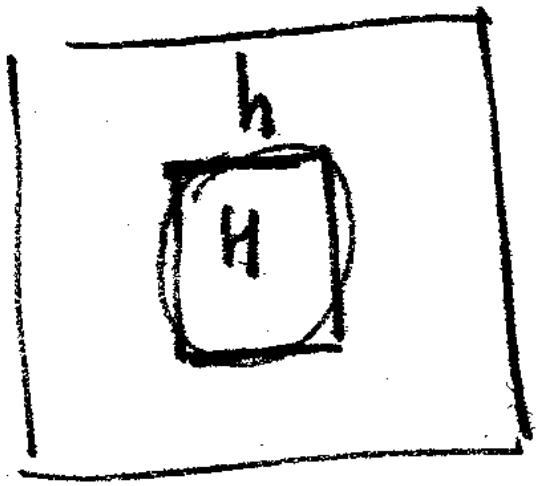
Typeset sketch; limitations are recorded in the note.

Kac, Helfand. Exact, in explicit form, for some cases [source endings: точн. в яв. виду для нек-рых случаях].



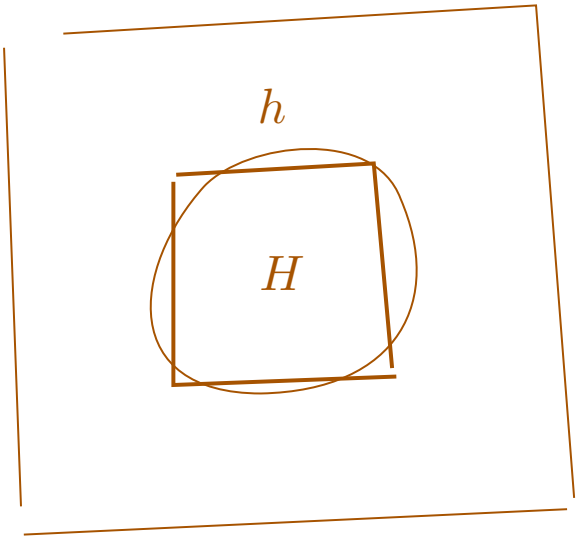
Original source details: GELFAND-P000004

Uncertain reading. After another enlargement, the line reads “точн. в яв. виду для нек-рых случаях”; the endings may instead be “виде” and “случаев”. The ungrammatical reading is not corrected automatically.



Original source details: GELFAND-P000004

Diagram and typed description. A large square with a smaller outlined region inside; H at its center and h above the smaller region. The small contour is partly square and partly rounded; its precise boundary has not been reconstructed.



Typeset sketch; limitations are recorded in the note.

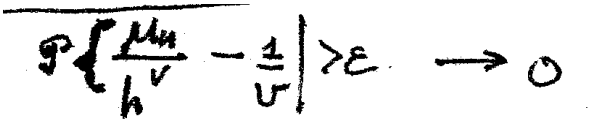
$$\frac{\mu_H}{h^\nu}.$$

μ_H is the number of particles in H — a subcube. ν is the dimension.

$$\frac{\mu_H}{h^\nu} = \frac{1}{v} \qquad \text{as in the whole space.}$$

Law of large numbers.

$$\mathsf{P}\left\{\left|\frac{\mu_H}{h^\nu} - \frac{1}{v}\right| > \varepsilon\right\} \longrightarrow 0.$$



Original source details: GELFAND-P000004

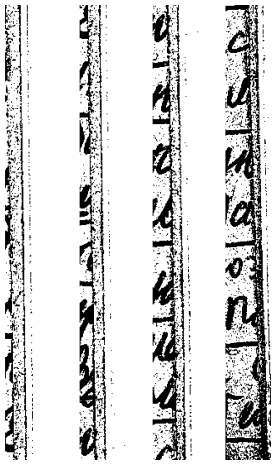
Uncertain reading. In the probability formula the right vertical bar before $> \varepsilon$ is visible; a left bar before the fraction has not been read confidently. No matching bar is silently supplied.

$$\ell \to \infty, \qquad H \to \infty, \qquad d < \frac{h}{\ell} < 1 - d, \qquad d > 0.$$

Uncertain reading. At the lower right edge a small sign resembles “2” or a broken-off letter; it has not been adopted as a page number.



Original source details: GELFAND-P000004

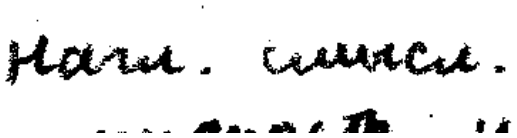


Original source details: GELFAND-P000004

Undeciphered passage. Along the extreme left edge is a narrow strip of neighboring handwritten text, cropped to isolated endings and strokes; no connected text can be read.

GELFAND-P000005

If this does not hold, a phase transition takes place.

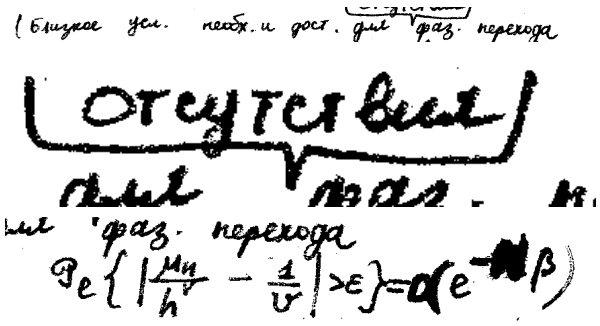


Original source details: GELFAND-P000005

Uncertain reading. Нагл. [candidate abbreviation; the middle strokes are not fully distinguishable]

meaning. 2 phases coexist; the particles are distributed unevenly — — liquid and gas. [Above the line:] (law of large numbers). [Inserted above “phase transition”, with a bracket:] absence. (A closely related condition is necessary and sufficient for a phase transition

$$\mathsf{P}_\ell\left\{\left|\frac{\mu_H}{h^\nu}-\frac{1}{v}\right|>\varepsilon\right\}=O(e^{-N\beta}).$$



Original source details: GELFAND-P000005

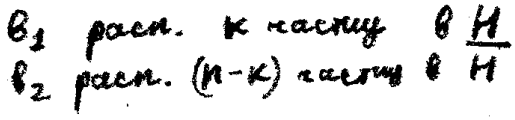
Uncertain reading. The exponent of e contains a heavily written sign before β; the candidate is $-N\beta$, but N is obscured by overlapping strokes. The parenthesis opened before “A closely related” is not clearly closed in the scan.

[At right:] N is the number of particles.

Calculate the probability that

$$\mathsf{P}\{\mu_H=k\}=Z^{-1}\sum_{\theta\in\mathcal{B}_k}\exp\{-U(\theta)/T\}.$$

Over configurations in which H contains k particles. θ is a configuration of N particles. $\theta\in\mathcal{B}_k$. Then $\theta=(\theta_1,\theta_2)$. θ_1 is a configuration of k particles in H. θ_2 is a configuration of $(N-k)$ particles in $\overline{H}$.

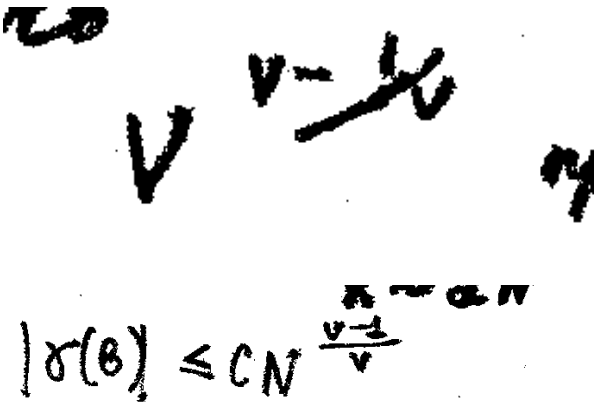


Original source details: GELFAND-P000005

Uncertain reading. The upper bar on H in the line about θ_2 is not completely distinct; the candidate is $\overline{H}$.

$$U(\theta) = U(\theta_1) + U(\theta_2) + \gamma(\theta).$$

$U(\theta_1)$ is the interaction inside. $U(\theta_2)$: neighbors outside. $\gamma(\theta)$ is the interaction across the surface. $\gamma(\theta)$ can be neglected asymptotically. (Across the boundary: of the order of the number of boundary points) $V^{\nu-1/\nu}$ as $N \to \infty$. $k \sim dN$.

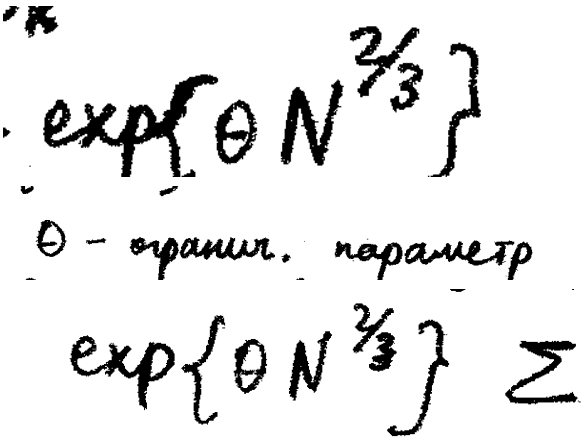


Uncertain reading. The exponent on V looks like $\nu - 1/\nu$, without a confidently distinguishable common numerator. It has not been changed automatically to $(\nu - 1)/\nu$.

Original source details: GELFAND-P000005

$$|\gamma(\theta)| \le CN^{\frac{\nu-1}{\nu}}.$$

$$\sum \exp\{-U(\theta)/T\} = \sum_{\theta \in \mathcal{B}_k} \exp\{-U(\theta_1)/T\} \exp\{-U(\theta_2)/T\}$$
$$\times \exp\{\theta N^{2/3}\} =$$



Uncertain reading. The letter before $N^{2/3}$ resembles θ or Θ; below, the same letter is glossed as “bounded parameter”. It may differ from the configuration sign.

Original source details: GELFAND-P000005

θ is a bounded parameter.

$$= \exp\{\theta N^{2/3}\} \sum_{\theta_1} \exp\{-U(\theta_1)/T\} \sum_{\theta_2} \exp\{-U(\theta_2)/T\}.$$

[Beneath the first sum:] configurations.

GELFAND-P000006

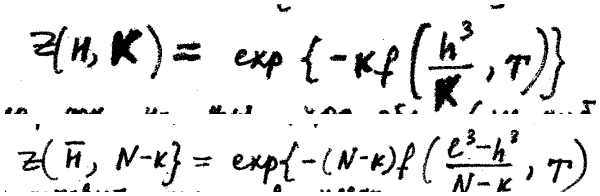
Factor 1: Z, calculated for the subcube. Factor 2 — ditto — Z for the outer cube.

$$\mathsf{P}\{\mu_H = k\} = \frac{Z(H,k)Z(\overline{H}, N-k)}{Z(V,N)} \exp\{\theta N^{2/3}\}.$$

[Asterisk.] Apply to them the asymptotic formula

$$Z(V,N) \approx \exp\{-Nf(v,T)\},$$

$$Z(H,k) = \exp\{-kf(h^3/k,T)\}.$$



Uncertain reading. The numerator of the argument of f in this line visibly has exponent 3 on h, although ν is used later. The written h^3 is retained.

Original source details: GELFAND-P000006

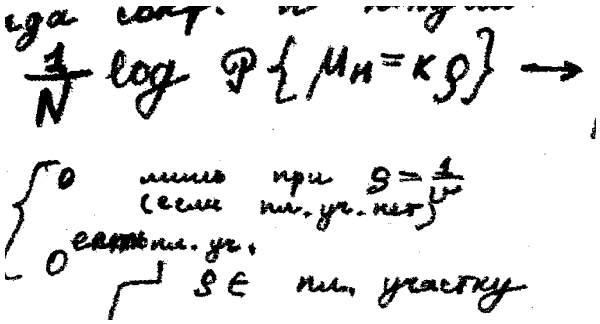
It can also be proved for well-behaved regions (not a cube). The limiting formula does not depend on the region's shape.

$$Z(\overline{H},N-k) = \exp\{-(N-k)f((\ell^3-h^3)/(N-k),T)\}.$$

Substitute there under the assumption that f has a linear segment. If f has a linear segment (the derivative has a horizontal one), then terms cancel and we obtain

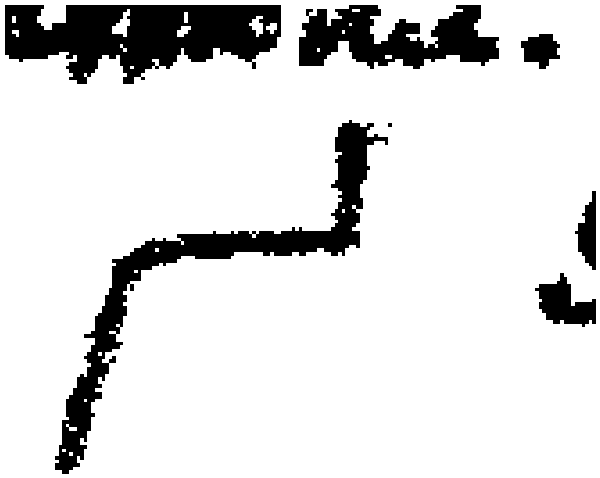
$$\frac{1}{N}\log \mathsf{P}\{\mu_H = k\rho\} \longrightarrow \begin{cases} 0 & \text{only at } \rho = 1/v \text{ (if there is no linear segment)}, \\ 0 & \rho \in \text{ the linear segment.} \end{cases}$$

[Cancelled by the second branch:]



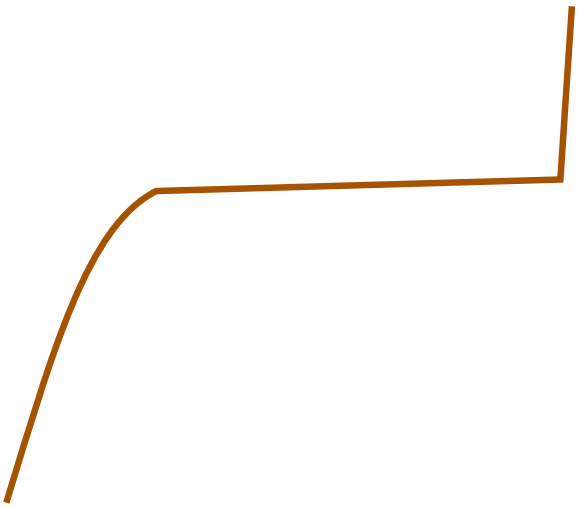
Uncertain reading. The event's argument looks like $\mu_H = k\rho$, but the final sign may be θ or δ. The density letter at right resembles ρ. The cancelled text above the second branch has not been read.

Original source details: GELFAND-P000006



Original source details: GELFAND-P000006

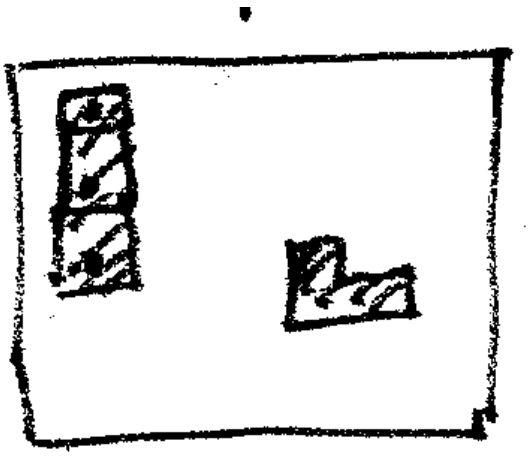
Diagram and typed description. At right of the second branch of the limit is a small stroke: from the lower left, a line rises almost vertically, then runs almost horizontally to the right, and rises again at the right end. Separate axes, arrows and labels have not been established; the sketch's mathematical meaning requires the original image.



Visible stroke; no axes or values supplied.

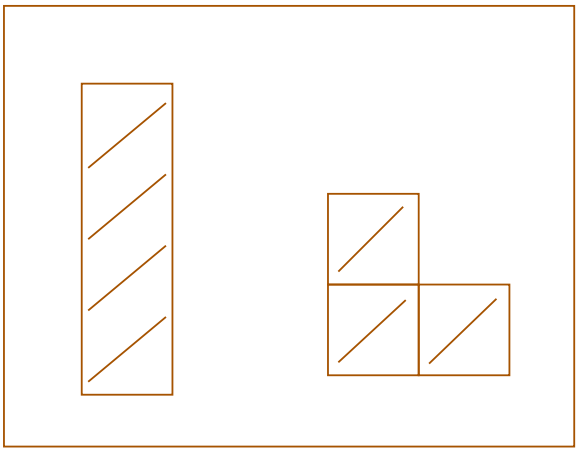
$\mathsf{P} \to 0$ exponentially. ($1/v$ follows from the convexity of f.)

Arrangement of particles. Its boundary; around each particle a cube of side 1. $\Gamma'(\theta)$ is the boundary of the occupied region. $\Gamma(\theta)$ is the length [inserted below:] area.



Original source details: GELFAND-P000006

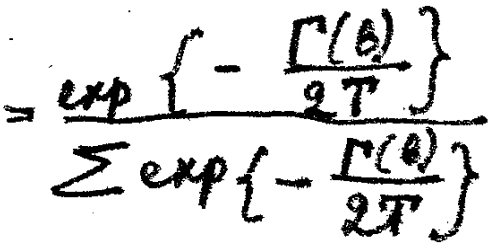
Diagram and typed description. Inside a square frame, a vertical shaded chain is at left; its internal divisions merge with the hatching. At right, three cells form a corner: two below and one above the left cell. The left cell count is unresolved.



Partial left-chain outline; internal divisions unresolved. Three-cell corner at right.

Replace the Gibbs probability.

$$\mathsf{P}(\theta) = \frac{\exp\{-U(\theta)/T\}}{\sum \exp\{-U(\theta)/T\}} = \frac{\exp\{-\Gamma'(\theta)/(2T)\}}{\sum \exp\{-\Gamma'(\theta)/(2T)\}}.$$

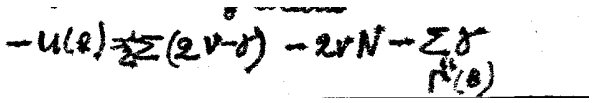


Original source details: GELFAND-P000006

Uncertain reading. The prime on Γ in the two exponents after the equality is not completely clear; the visible letter resembles Γ' or Γ.

[At right:] $2\nu - \delta$. Unoccupied squares $\Gamma'(\theta)$. [Inserted:] at neighbors.

$$-U(\theta) \geq \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta.$$



Original source details: GELFAND-P000006

Uncertain reading. The sign after $-U(\theta)$ is written over another sign: candidates are $\geq$, $\leq$, and $=$. The sign before the final sum and the sum's index may belong to another term; the visible candidate is retained without a mathematical correction.

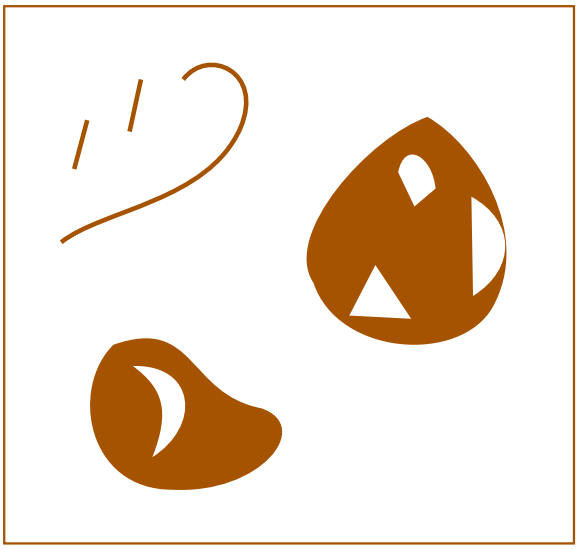
GELFAND-P000007

($U(\theta)$ is minus the number of neighbors.) A smaller boundary makes an arrangement more favorable. Force them into one corner, but this is unfavorable (there are few such arrangements in all). A finite number of clumps. (Inside the gas: liquid.)



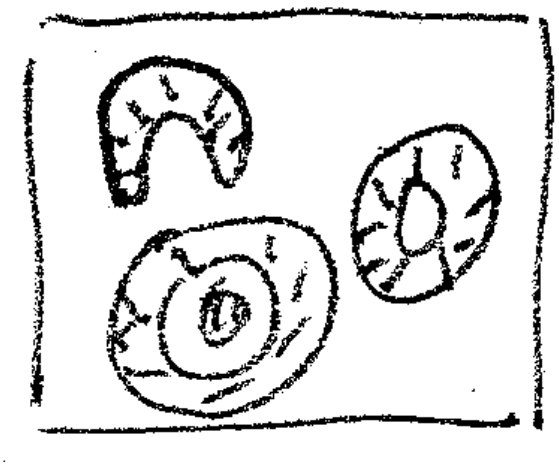
Original source details: GELFAND-P000007

Diagram and typed description. Small square drawing at right: two filled regions and small strokes in the upper left corner. The number and meaning of the small strokes are unclear.



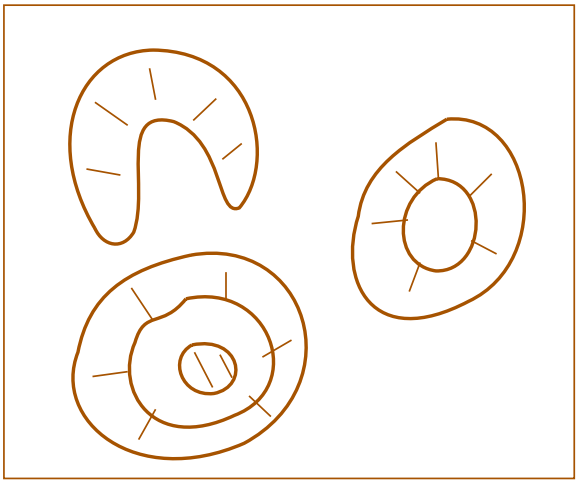
Partial sketch of patches and visible marks; small strokes remain uninterpreted.

And this contradicts the law of large numbers.



Original source details: GELFAND-P000007

Diagram and typed description. A large rectangular frame at left contains three separate rounded regions with inner contours and shaded parts; one has a central island. The precise cycles, shading and nesting require the original image.

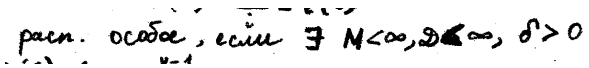


Nesting sketch, retaining the lower opening of the upper-left group.

The whole boundary splits into connected cycles. Each arrangement has its own number. Number them

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots, \qquad \Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

An arrangement is special if $\exists N < \infty$, $D < \infty$, $\delta > 0$.

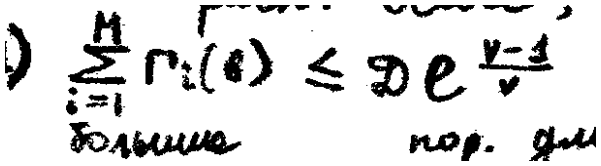


Original source details: GELFAND-P000007

Uncertain reading. In this line the first finite parameter resembles N, whereas the later summation limit resembles M. This difference is not smoothed out.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\frac{\nu-1}{\nu}}.$$

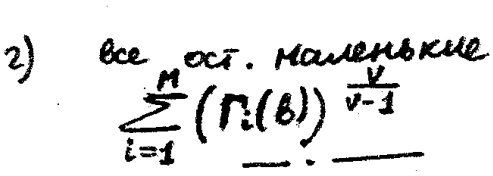
[Beneath the sum:] Large. Of the order of the boundary length. (Almost the whole cube.)



Original source details: GELFAND-P000007

Uncertain reading. The exponent on ℓ in the first condition is written as a fraction, read as $(\nu-1)/\nu$; in GELFAND-P000010 the analogous line has $\nu - 1$. The difference is retained.

$$2) \quad \text{all the others small} \qquad \sum_{i=1}^{M} \big(\Gamma_i(\theta)\big)^{\frac{\nu}{\nu-1}}.$$



Original source details: GELFAND-P000007

Uncertain reading. The visible limits of the second sum look like 1 and M, although the caption says “all the others small”. No condition is visible after the sum; none is supplied from the later scan.

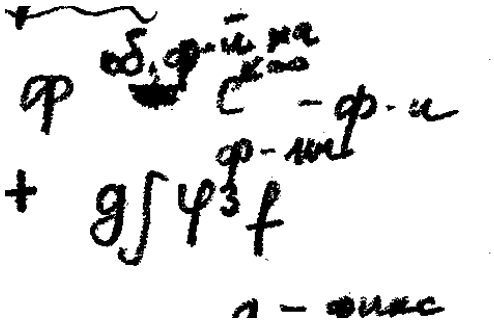
[Dividing line.]

24/IX. Thursday. 16 h, room 13-11. Prof. Segal (USA). Theory of functions in infinite-dimensional space.

$$+\Box\varphi = m^2\varphi + g\varphi^3.$$

$$F_f(\varphi) = \int (-\Box + m^2) f\varphi + g \int \varphi^3 f.$$

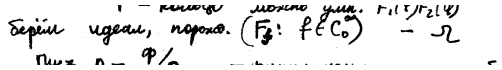
[Above φ^3:] φ — мн. [source abbreviation]. φ [arrow] C_0^∞ — functions.



Original source details: GELFAND-P000007

Uncertain reading. Above the note at right are the visible words “б, ф-й на” [fragment: functions on] and an arrow toward C_0^∞; this may define the function φ on the space of test functions, but no text is supplied.

g is fixed. $F_f \in \Phi$ is a nonlinear functional of φ. Φ is a ring; one can multiply $F_1(\varphi)F_2(\varphi)$. Take the ideal generated by $(F_f : f \in C_0^\infty)$ — Ω.



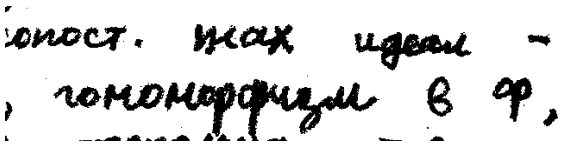
Original source details: GELFAND-P000007

Uncertain reading. The index on F in the ideal’s definition resembles f; the source sign after the parenthesis is Ω or a calligraphic J.

Let $p = \Phi/\Omega$ be the function ring on the manifold of solutions of the given partial differential equation. To obtain a solution $(1) \stackrel{\text{def}}{=}$ a point of the space of maximal ideals of the ring p. An ordinary solution φ_0 [a check mark above the word] is assigned a maximal ideal: consider the map

$$F \longrightarrow F(\varphi_0),$$

a homomorphism into Φ, F_f is zero on the ideal Ω. This is a homomorphism of the quotient ring as well, i.e. a maximal ideal.



Original source details: GELFAND-P000007

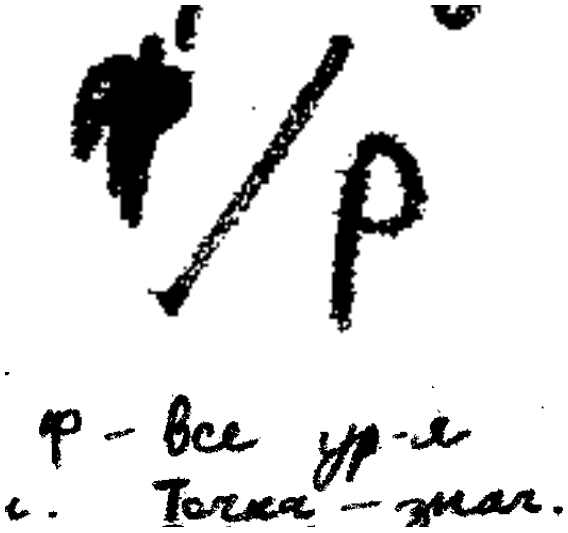
Uncertain reading. After “a homomorphism into”, the visible letter resembles Φ, although its meaning might suggest a field. The letter candidate is retained rather than corrected mathematically.

GELFAND-P000008

A maximal ideal is a generalized solution of a nonlinear equation. The algebraic case.

$$f_1(x_1, \ldots, x_n) = 0, \qquad f_k(x_1, \ldots, x_n) = 0.$$

Solutions in terms of a ring: in the ring of all polynomials. The ideal generated by these functions is p, and the quotient ring is Φ/p.

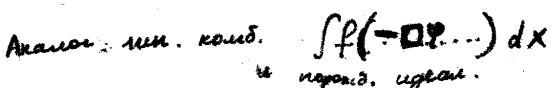


Original source details: GELFAND-P000008

Uncertain reading. In the fraction, the numerator resembles Φ, P, or a heavily written A; the candidate Φ is retained.

Φ — all equations. This is the ring of algebraic functions. A point, the value at the point, is precisely a maximal ideal. In the algebraic case, conversely as well. Analogy: linear oscillations.

$$\int f(-\Box\varphi\ldots)\,dx.$$

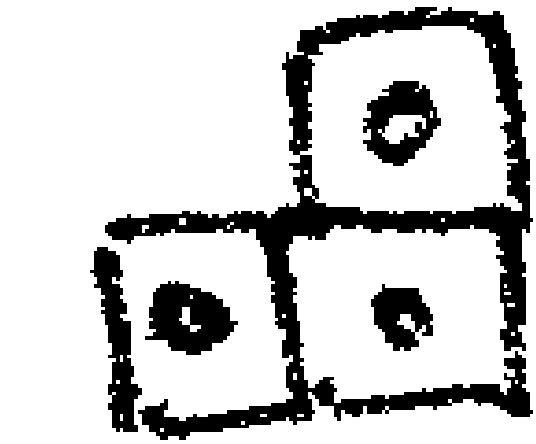


Original source details: GELFAND-P000008

Uncertain reading. The part of the integral after $-\Box$ looks like $\varphi\ldots$, but the letter is partly inked over.

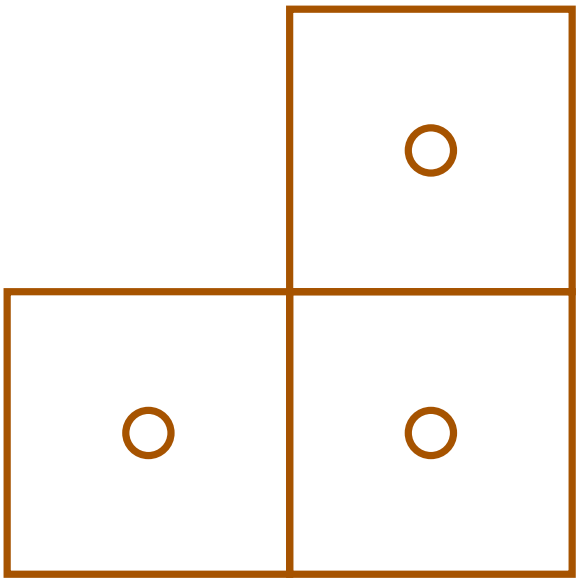
And the corresponding ideal.

Dobrushin. $\nu = 2, 3$. $V_\ell = \ell^\nu$, $N < V_\ell$, $\theta = (x_1, \ldots, x_N)$.



Original source details: GELFAND-P000008

Diagram and typed description. Original and existing typeset recreation.



$\Gamma(\theta)$ is the boundary length. $U(\theta)$ is the number of neighbors (pairs of neighbors). The Gibbs distribution

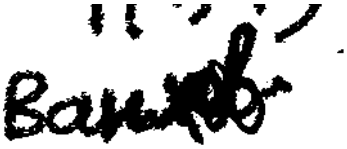
$$p(\theta) = \frac{\exp\{U(\theta)/T\}}{\sum\limits_{\theta\in\mathcal{B}(V_\ell,N)} \exp\{U(\theta)/T\}} = \frac{\exp\{-\Gamma(\theta)/(2T)\}}{\sum\limits_{\theta} \exp\{-\Gamma(\theta)/(2T)\}}.$$

[Beneath the first sum:] all arrangements.

$$(\text{since } 2U(\theta) = 2\nu N - \Gamma(\theta)).$$

Phase transition. Let the lattice grow, $\ell \to \infty$, and let $N_\ell \to \infty$. The limit $V_\ell / N_\ell \to v$ is the specific volume. It turns out that

$$\frac{1}{N} \log \sum_\theta \exp\{U(\theta)/T\} \longrightarrow f(v, T).$$



Uncertain reading. Beneath the limit the name reads “Ван Хов” or “Ванхов” [Van Hove]; the middle and ending are heavily overwritten.

Original source details: GELFAND-P000008

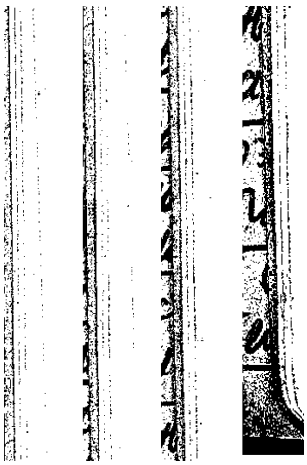
(v, T) is called a point of phase transition if

$$\frac{\partial f(v,T)}{\partial v} = 0 \quad \text{in some neighborhood at } T = \text{const.}$$



Uncertain reading. Between T and const, the sign resembles $=$ or $\neq$; the candidate $=$ is retained, but the reading is not asserted.

Original source details: GELFAND-P000008



Undeciphered passage. At the extreme left edge is a narrow cropped column of handwritten endings from neighboring text; continuous words cannot be recovered.

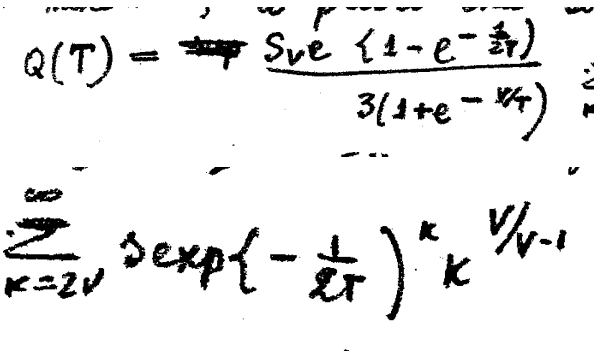
Original source details: GELFAND-P000008

GELFAND-P000009

(Since the function is linear; a priori it is convex.)

$$Q(T) = \frac{S_\nu e(1 - e^{-1/(2T)})}{3(1 + e^{-\nu/T})} \sum_{k=2\nu}^{\infty} 3 \exp\{-1/(2T)\}^k k^{\frac{\nu}{\nu-1}}.$$

[After the first equality, a cancelled fragment:]

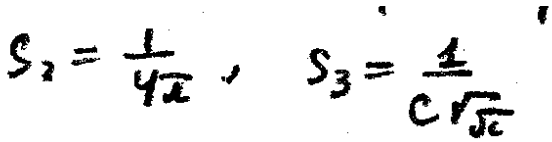


Uncertain reading. The cancelled sign after $=$ is illegible. In the numerator, the factor after S_ν resembles e; the upper summation limit resembles ∞ or a double stroke. The exponent k may apply to the factor 3 as well, but in the scan it stands by the closing brace. The variants are not replaced with a standard formula.

Original source details: GELFAND-P000009

S_ν is the volume of a ν-dimensional sphere with surface 1.

$$S_2 = \frac{1}{4\pi}, \qquad S_3 = \frac{1}{C\sqrt{\pi}}.$$



Uncertain reading. The first factor in the denominator of S_3 resembles C, c, or 6; the written C is only a candidate.

Original source details: GELFAND-P000009

If

$$Q(T) < \frac{1}{v} < 1 - Q(T),$$

then a phase transition takes place.

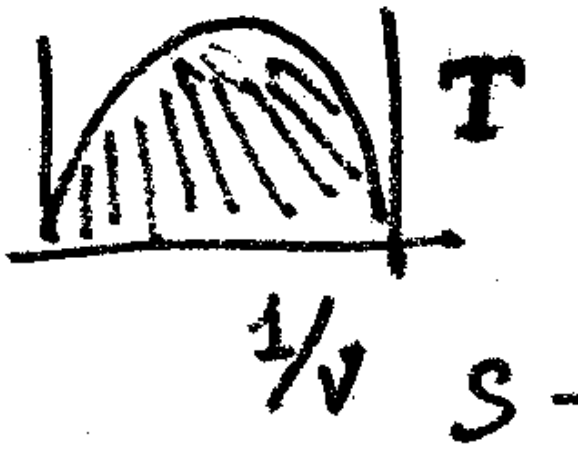
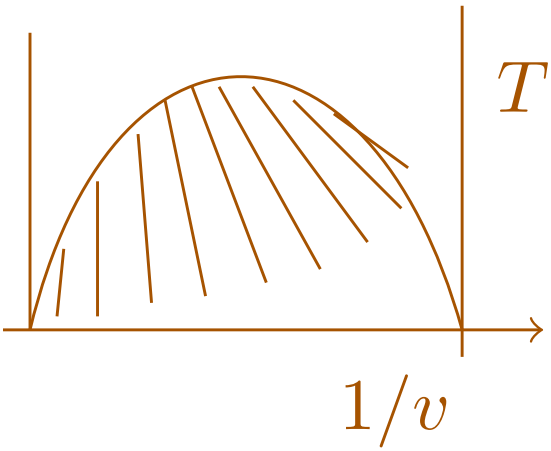


Diagram and typed description. Original and existing typeset recreation.



Original source details: GELFAND-P000009

[Dividing line.]

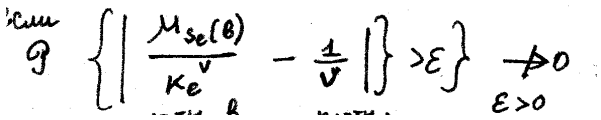
S is a subcube. $S_\ell \subset V_\ell$. K_ℓ is the side of S_ℓ.

$$0 < d < \frac{K_\ell}{\ell} < 1 - d.$$

Introduce a random variable. $\mu(S_\ell)$ is the number of points of the arrangement θ that fall in S_ℓ. $\mu_{S_\ell}(\theta)$. If

$$\mathsf{P}\left\{\left|\frac{\mu_{S_\ell}(\theta)}{K_\ell^\nu} - \frac{1}{v}\right|\right\} > \varepsilon\} \not\longrightarrow 0, \qquad \varepsilon > 0,$$

then there is a phase transition.



Original source details: GELFAND-P000009

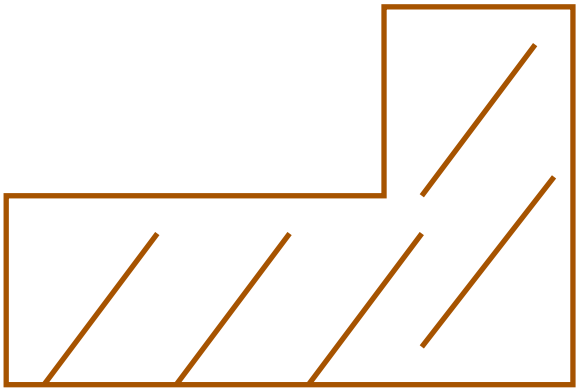
Uncertain reading. After the right absolute-value bar, the scan shows a brace before $> \varepsilon$ and another after ε. This inconsistent notation is retained; the usual event candidate has the absolute value greater than ε inside a single pair of braces.

[Beneath the first fraction:] density in the subcube. [Beneath $1/v$:] density in the whole cube. And a closely related condition is necessary. (Physical meaning: coexistence of 2 phases.)



Original source details: GELFAND-P000009

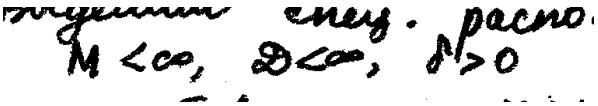
Diagram and typed description. Original and existing typeset recreation.



The boundary splits into components — — but a connection through corners ($\nu = 2$) or an edge ($\nu = 3$) is not counted.

$$\Gamma(\theta) = \sum_i \Gamma_i(\theta).$$

$\Gamma_i(\theta)$ is the length or area of a boundary component. Definition. Single out a special arrangement. It depends on 3 numbers, $N < \infty$, $D < \infty$, $\delta > 0$.



Original source details: GELFAND-P000009

Uncertain reading. The first letter resembles N, but admits M; the enlarged fragment does not resolve the distinction. It is not automatically reconciled with the upper summation limit in the next frame.

Number the cycles in decreasing order of length

$$\Gamma_1(\theta) \geq \Gamma_2(\theta) \geq \cdots .$$

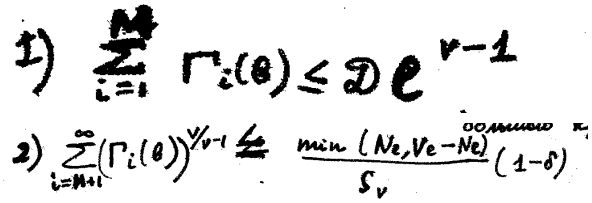
GELFAND-P000010

Conditions.

$$1) \quad \sum_{i=1}^{M} \Gamma_i(\theta) \leq D\ell^{\nu-1}.$$

(Of the order of the surface of the large cube.)

$$2) \quad \sum_{i=M+1}^{\infty} \big(\Gamma_i(\theta)\big)^{\frac{\nu}{\nu-1}} \leq \frac{\min(N_\ell, V_\ell - N_\ell)}{S_\nu}(1-\delta).$$

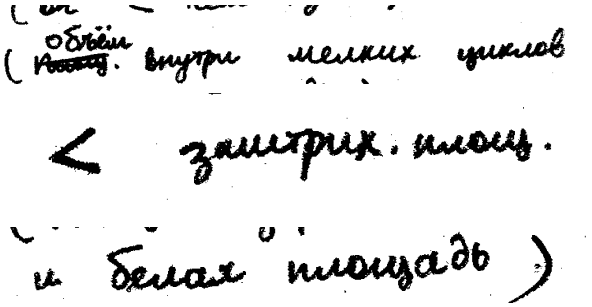


Original source details: GELFAND-P000010

Uncertain reading. The inequality sign before the fraction is heavily overwritten: candidates are $\leq$ and $<$. The parameter in the upper limit of the first sum resembles M or N.

$$S_\nu \big(\Gamma_i(\theta)\big)^{\frac{\nu}{\nu-1}}$$

Area of a circle with length Γ_i. A ball has maximal volume; this is an upper bound on the volume of the region inside the cycles. (It is $<$ what is needed to contain all occupied points.) ([A word is cancelled; inserted above:] volume. Inside small cycles $<$ the shaded area and the white area.)



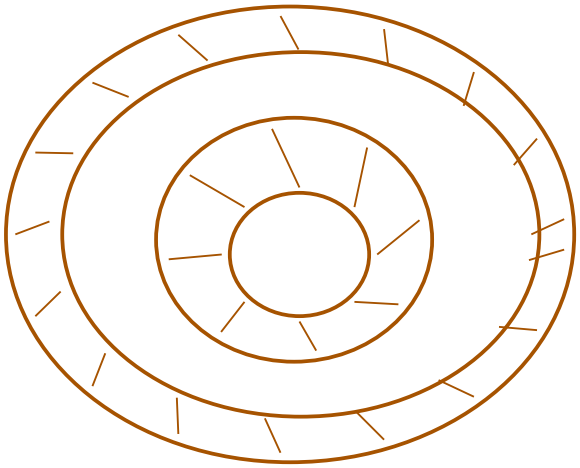
Original source details: GELFAND-P000010

Uncertain reading. The word beneath the inserted “volume” is heavily cancelled and illegible. “Area” in the two lines is retained as in the scan, without replacing it with volume.



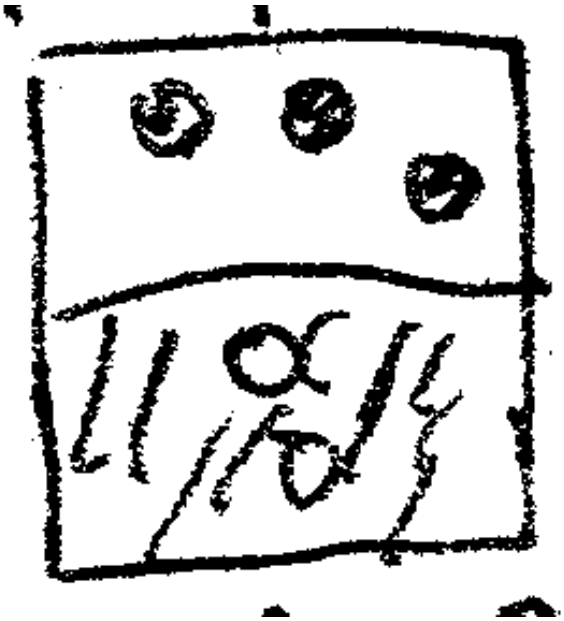
Original source details: GELFAND-P000010

Diagram and typed description. Four nested rounded boundaries: an outer shaded ring, a white gap and an inner ring with a white centre. Individual pen irregularities and merged strokes remain visible in the original.



Four boundaries and two shaded rings; schematic scale.

Neither all the points nor all the empty sites can be forced into it. An example of an arrangement where the conditions do not hold.



Original source details: GELFAND-P000010

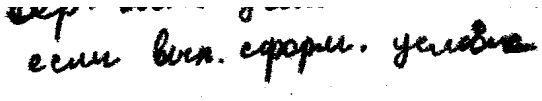
Diagram and typed description. A rectangle is divided by an almost horizontal curve: three small rounded regions above, shading and several rounded unshaded regions below. The precise boundaries and number of lower regions are uncertain.



Partial sketch: the uncertain lower contour remains open.

This is a phase transition. Everything reduces to 2 lemmas.

1) $\exists M, D, \delta$ such that the probability that the conditions hold $\longrightarrow 1$, if the stated condition holds. [Beneath the parameters:] independent of everything.

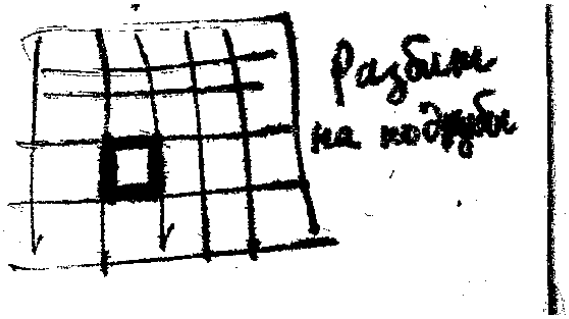


Original source details: GELFAND-P000010

Uncertain reading. In the line after “if ... holds”, the word looks like “сформ.” [stated] or “строгое” [strict]; nothing is supplied from the meaning.

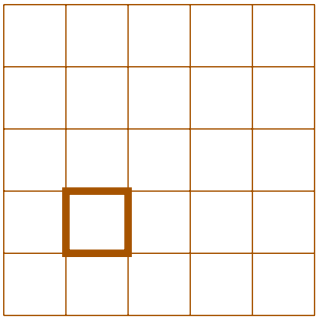
2) Lemma 2. If almost all arrangements are special, then there is a phase transition. Idea of the proof of Lemma 2. G is the volume of the region enclosed by the small cycles.

$$G \leq N_\ell(1-\delta), \qquad \frac{G}{V_\ell} \leq \frac{1}{v}(1-\delta).$$



Original source details: GELFAND-P000010

Diagram and typed description. A five-by-five rectangular grid; the second cell from the left in the fourth row from the top is emphasized. Caption: “Divided into subcubes”.

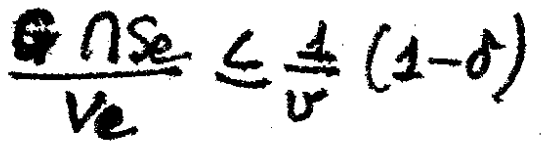


Divided into subcubes

Five-by-five grid; cell (2,4) counted from left and top is emphasized.

There exists a subcube S_ℓ:

$$\frac{G \cap S_\ell}{V_\ell} \leq \frac{1}{v}(1-\delta).$$

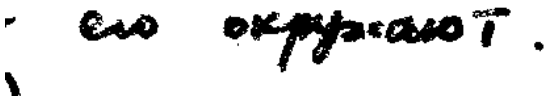


Uncertain reading. In the numerator, the heavy sign before $\cap S_\ell$ resembles G; the denominator clearly resembles V_ℓ. The candidate is retained; neither a measure of the intersection nor the subcube's volume is supplied from the meaning.

Original source details: GELFAND-P000010

GELFAND-P000011

Let all large cycles



Uncertain reading. surround co / ∞. [The two-loop sign before the verb is uncertain.]

Original source details: GELFAND-P000011

.

2 cases.



Diagram and typed description. Two small rectangles: the left contains four filled circles; the right contains four round marks, three with visible light centres, while the lower right one is darker and more heavily outlined. Additional small, partly curved strokes lie inside the right rectangle. Their meaning and exact directions are not established.



Partial sketch: small interior strokes remain in the original.

Original source details: GELFAND-P000011

Lemma.



Uncertain reading. Непрс. [uncertain source abbreviation]

Original source details: GELFAND-P000011

a boundary of k [cancelled:



Original source details: GELFAND-P000011

Uncertain reading. точк. [points; uncertain cancelled reading]

; inserted above/below: links] $\leq 3^k$.

Gain 3, loss.



Original source details: GELFAND-P000011

Diagram and typed description. Beneath the words about gain and loss is a horizontal line with short end strokes and a vertical segment extending downward. This is the source sketch; no labels have been read confidently.



Visible strokes, with no supplied labels or arrows.

$$\exp\left\{-\frac{1}{2T}\right\}, \qquad \left(3\exp\left\{-\frac{1}{2T}\right\}\right)^k.$$

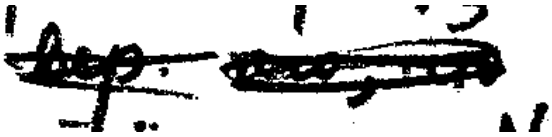
Problem. (unsolved) Let N be large. Turn with probability $\frac{1}{3}$, but returning is forbidden [cancelled:



Original source details: GELFAND-P000011

Uncertain reading. не [not; uncertain candidate for a short word]

]. What is [cancelled:

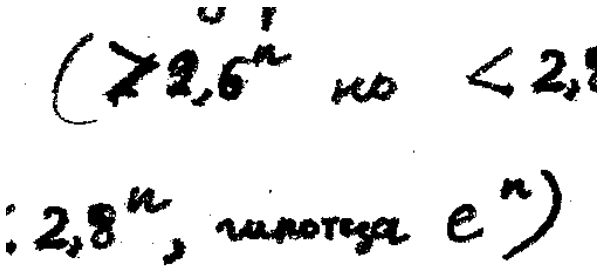


Original source details: GELFAND-P000011

Uncertain reading. вер. ... [prob.; candidate for the first abbreviation; the following cancelled writing is unreadable]

] the expected distance he will travel in N steps?

(If returning is allowed, then $\sim \sqrt{n}$.) If returning is allowed, then



Uncertain reading.
($\geq 2{,}6^n$ but $< 2{,}8^n$, conjecture e^n); the first sign may also read $>$; additional strokes are uncertain

Original source details: GELFAND-P000011

.

[Horizontal divider.]

28/IX Segal. Theory of functions on the set of solutions of a relativistic differential equation. [In the left margin: They are all written this way.]

$$u' = Au + K(u) \qquad (A) \qquad u(t), \quad -\infty < t < +\infty,$$

$$A = -A^*, \qquad u \in L^2, \qquad u(t) \in H.$$

K is a given nonlinear operator. In integral form

$$u(t) = e^{A(t-t_0)}u(t_0) + \int_{t_0}^{t} e^{A(t-s)}K(u(s))\,ds \qquad (B).$$

When $K \in C^1$, i.e. $\partial_u K = K_u$ exists in the Fréchet sense and is continuous, every solution of (B) is a solution of A (if $u(t)$ is in the domain of A).

[Horizontal divider.]

Suppose (B) has a global solution for every initial datum at t_0 (u_0). (And $K \in C^1$.) In quantum mechanics

$$\Box\varphi = f(\varphi)$$

[Beneath □: d'Alembertian.]

$$u(t) = [\varphi(\cdot, t), \dot{\varphi}(\cdot, t)], \qquad \varphi = \varphi(\vec{x}, t).$$

K is a nonlinear operator,



Uncertain reading. Немыц. / Намыц. [Nemyts. / Namyts.; the final source abbreviation is uncertain]

Original source details: GELFAND-P000011

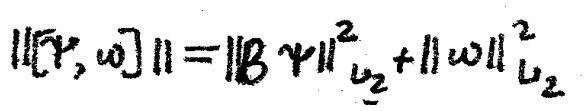
.

GELFAND-P000012

$$\|[\varphi, \omega]\| \qquad \Box\varphi = m^2\varphi + f(\varphi)$$

[Beneath the formula: A shorter form.]

$$B = \sqrt{m^2 I - \Delta}, \qquad \|[\varphi, \omega]\| = \|B\varphi\|_{L^2}^2 + \|\omega\|_{L^2}^2.$$



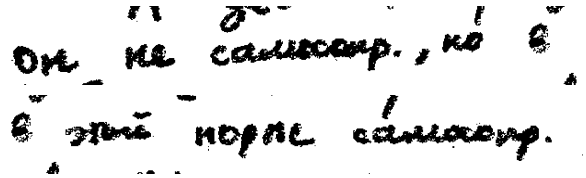
Original source details: GELFAND-P000012

Uncertain reading. No square is visible above the norm on the left of the last formula; the right side contains squares. The written form is retained without mathematical correction.

A is given by a matrix of order 2,

$$\begin{pmatrix} 0 & I \\ -B^2 & 0 \end{pmatrix}.$$

It is not self-adjoint, but in this norm it is self-adjoint.



Original source details: GELFAND-P000012

Uncertain reading. The source abbreviation “самосопр.” [self-adjoint] is read this way in both places; the possible abbreviation and its agreement with the preceding $A = -A^*$ require comparison.

Any equation of relativistic physics can be written this way.

M is the collection of all solutions.

$$M = [u(\cdot);\ (**) \text{ satisfies}]$$

[At right/beneath the formula: of t;



Original source details: GELFAND-P000012

Uncertain reading. различные B. [various B.; uncertain reading of the underlined note]

] Now let $K \in C^\infty$. M is a Banach manifold (every point has a neighbourhood homeomorphic to a Banach [cancelled:



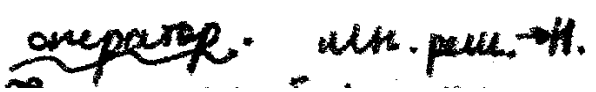
Original source details: GELFAND-P000012

Uncertain reading. unreadable word

] space; the transition is a C^∞ function in the Fréchet sense).

$$S_t : M \longrightarrow H, \qquad u(\cdot) \longmapsto u(t).$$

[At right: operator.



Original source details: GELFAND-P000012

Uncertain reading. мн. реш. $\longrightarrow H$ / им. реш. $\longrightarrow H$ [set of sol. / им. sol.; the first source abbreviation is uncertain; the final arrow and H are visible]

] Theorem. There exists a unique C^∞ manifold structure for which the map S_t is a C^∞ isomorphism. $u(t, u_0)$ depends on u_0 in a C^∞ manner.

A true manifold. There is a tangent space $T_{u(\cdot)}$. The tangent map

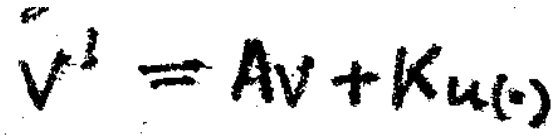
$$dS_t : T_{u(\cdot)} \longrightarrow T_{u(t)}.$$

Of course, at every point it is isomorphic to the space itself.

$$\lambda \in T_{u(\cdot)}, \qquad dS_t(\lambda) = v(t) \in H.$$

Thus a function corresponds to each tangent vector. [Underlined and cancelled: Lemma.] Denote by $v(t)$ (not the value of v at the point t); variational equation, an equation of first order:

$$v' = Av + K_{u(\cdot)}.$$



Uncertain reading. No separate factor v is visible after $K_{u(\cdot)}$ on this line; the written form is retained.

Original source details: GELFAND-P000012

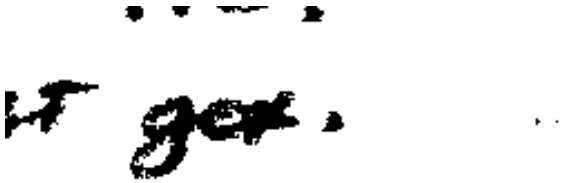
Solutions of this equation arise from tangent vectors λ, and conversely.

1) Canonical differential form on M.

2) Poisson brackets (analogy with the finite-dimensional case).

GELFAND-P000013

3) Vector fields in M. In their terms there will be



Uncertain reading. зд. [the final source abbreviation is uncertain]

Original source details: GELFAND-P000013

.

4) Canonical commutation relations.

5) A representation and an explicit definition of the operators will be given.

5/X. Segal. I. Renormalized field equation. 1) Derivation. 2) Existence and uniqueness.

II. Definition of the scattering matrix S. A. Classical S operator. B. Extension of generalized probability distributions to the nonlinear case.

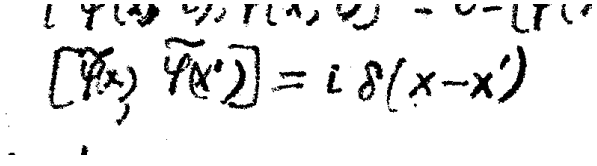
III. Discussion of interpretation in terms of particle theory.

[Horizontal divider.]

Consider the equation

$$\Box\widetilde{\varphi} = m^2\widetilde{\varphi} + \widetilde{\varphi}^3,$$

$$[\widetilde{\varphi}(\vec{x},t), \widetilde{\varphi}(\vec{x}',t)] = 0 = [\dot{\widetilde{\varphi}}(\vec{x},t), \dot{\widetilde{\varphi}}(\vec{x}',t)].$$



Uncertain reading. Candidates for the source line: $[\widetilde{\varphi}(x), \widetilde{\varphi}(x')] = i\delta'(x-x')$ / $[\widetilde{\varphi}(x), \dot{\widetilde{\varphi}}(x')] = i\delta(x-x')$. The dot above the second letter, the additional stroke after δ, and the exact placement of the arguments are not fully distinguishable.

Original source details: GELFAND-P000013

Loran Schwartz. There are 2 types of measures: Radon measures and simply measures [inserted above the last word: abstract]. For now, restrict to positive measures. An abstract measure on a set X, F is a σ-algebra. μ is a measure. μ is σ-additive on F.

[Horizontal line.]

Once a measure is given, one can consider the μ-measurable sets, F_μ on X, and integrate measurable functions.

$$f : X \longrightarrow F, \qquad \int_X f(x)\, d\mu(x).$$

[At right: values in a Banach space; F is a Banach space.]

[Horizontal line.]

Radon measure. X is an arbitrary locally compact space. $C(X)$ consists of continuous functions with compact supports, with the inductive-limit topology, and let $C'(X)$ be the dual space. Its elements are measures on X.

GELFAND-P000014

μ is positive if $\mu(\varphi) \geq 0$ for $\varphi \geq 0$. A measure is bounded if

$$|\mu(\varphi)| \leq C \max |\varphi|.$$

The 2nd definition is a bounded positive Radon measure. The same Lebesgue extension is performed with it. The integral also exists (with values in a Banach space).

The σ-field constructed from a Radon measure depends on μ, but there exists one independent of μ (substantially): the field of Borel sets (



Uncertain reading. min containing closed and open sets [the beginning of the parenthetical note is uncertain]

Original source details: GELFAND-P000014

).

[Horizontal divider.]

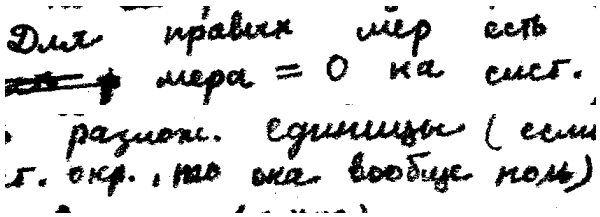
Why do various mathematicians prefer Radon measures? Radon measures exist only on locally compact spaces (a restriction in probability theory). But [cancelled:



Uncertain reading. two short unreadable words

Original source details: GELFAND-P000014

] Radon measures are more convenient.



Uncertain reading. For “right” measures there is разлож. / различ. единицы [decomp. / различ. of unity] (if [cancelled beginning] a measure $= 0$ on a system of neighbourhoods, then it is zero everywhere). [The first description of the measures and the abbreviation before “unity” are uncertain.]

Original source details: GELFAND-P000014

There are 5–15 theorems true for Radon measures (only).

$$X \xrightarrow{f} Y$$

[Beneath the arrow: topological spaces.] [Cancelled:

Uncertain reading. a short word before “Select”



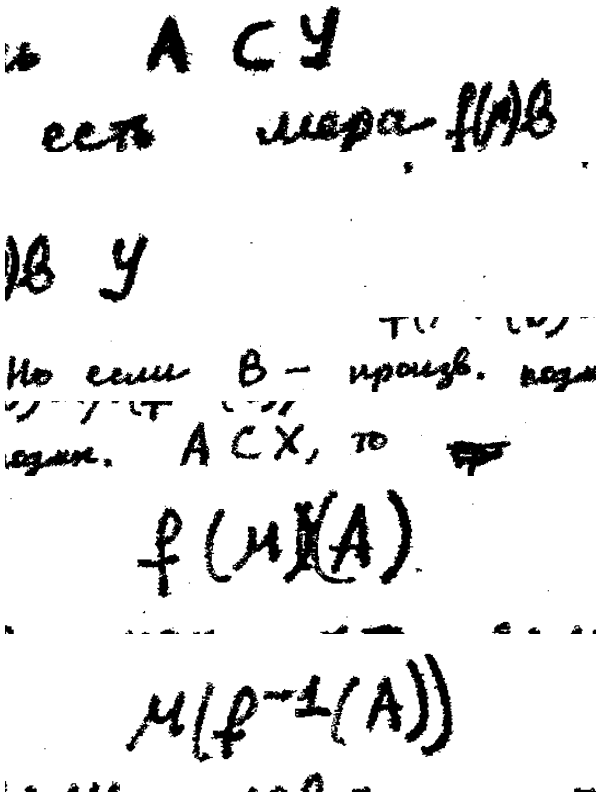
Original source details: GELFAND-P000014

.] Select the Borel field on X and on Y, and an abstract measure μ on X. Let $A \subset Y$. Then there is a measure $f(\mu)$ on Y:

$$f(\mu)(B) = \mu(f^{-1}(B)).$$

But if B is an arbitrary subset $A \subset X$, then [one sign cancelled]

$$f(\mu)(A) \qquad \mu(f^{-1}(A)).$$



Uncertain reading. The source alternates between A and B, and $A \subset X$ is read before the second pair of formulas; the letter designations have not been reconciled editorially.

Original source details: GELFAND-P000014

One can show that if the left side makes sense, the right side does too and they are equal; the converse is false. [Cancelled:



Uncertain reading. an unreadable beginning of a line

Original source details: GELFAND-P000014

.]

$$A \text{ — } f(\mu)\text{-measurable} \implies f^{-1}(A)\ \mu\text{-measurable}$$

If μ is a Radon measure, there is equivalence.

Example (contradictory)

$$X \subseteq Y = [0, 1]$$

such that X is a Lebesgue-nonmeasurable set, $Y \setminus X$ has inner measure 0 (or the outer measure of $X = 1$). A subset of X is Borel $\Longleftrightarrow$ it is an intersection with a Borel set of Y. μ is an abstract measure.

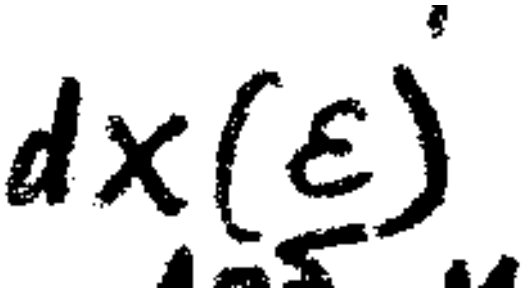
GELFAND-P000015

Definition. $\mu(D)$. $D \subset X$ is Borel.

$$D = X \cap E, \qquad E \text{ — Borel in } Y,$$

$$\mu(D) \underset{\text{def}}{=} dx(E)$$

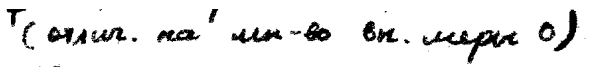
[Beneath $dx(E)$: Lebesgue measure.]



Original source details: GELFAND-P000015

Uncertain reading. In $dx(E)$ both letters are on the baseline; the case of the second letter is not fully distinguishable: $dx(E)$ / $dX(E)$. No subscript has been added.

The measure is the same (



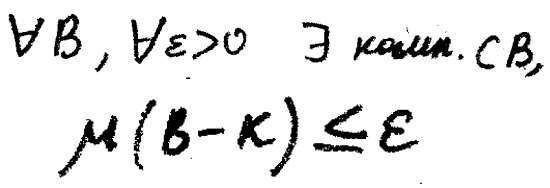
Original source details: GELFAND-P000015

Uncertain reading. отлич. на мн-во вн. меры 0 [differs on a set of вн. measure 0; uncertain parenthetical source note]

). We have defined an abstract measure on X. Its image [The following line repeats: its image] is dx. Take some (non-Borel) subset of $Y \setminus X$. It is nonmeasurable, but its inverse image is empty and hence measurable.

The aim of the talk is to show that local compactness is unnecessary. Let μ be a measure on F_B; then μ is a Radon measure if it is inner compact regular.

$$\mu \text{ Radon } \Longleftrightarrow \forall B,\ \forall \varepsilon > 0\ \exists \text{ compact } \subset B, \qquad \mu(b - k) \leq \varepsilon.$$



Original source details: GELFAND-P000015

Uncertain reading. Lowercase $b - k$ is read in the difference, although $\forall B$ precedes it. No separate set letter is visible between “compact” and $\subset B$; none has been supplied from the meaning.

[Horizontal divider.]

X is a topological space. Find a measure with all the properties of locally compact Radon measures. Suppose it is defined (set up) on Borel sets. A definition of a Radon measure valid beyond locally compact spaces. Let X be completely regular (such spaces can be embedded in compact spaces of maximal ideals).

$$X, \qquad \widehat{X} \text{ — Čech completion of } X$$

(maximal ideals of the algebra of continuous functions).

$$X \xrightarrow{i} \widehat{X}.$$

Complete regularity means that this is a homeomorphic map, rather than merely an injective map. Another definition of a Radon measure on X: a Radon measure on $\widehat{X}$ for which

$$\mu(\widehat{X} \setminus X) = 0.$$

One can show equivalence. In the locally compact case.

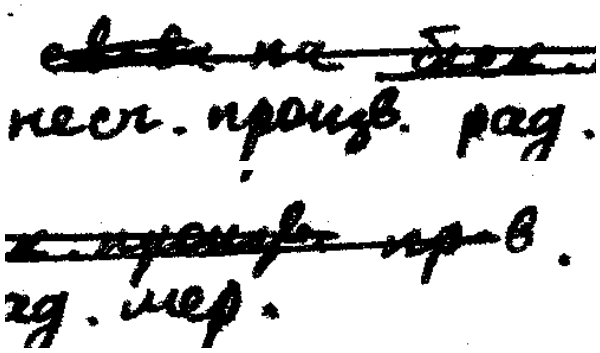
$$\mu \text{ Rad} \iff \forall B \text{ compact } \forall \varepsilon > 0 \; \exists \text{ compact} \quad \mu(B - k) < \varepsilon.$$



Original source details: GELFAND-P000015

Uncertain reading. The second sign in the difference resembles lowercase k; the first reads as B, but its case is uncertain ($B - k$ / $b - k$). No separate set letter or inclusion condition is visible after “compact”. The formula has not been supplemented from the preceding one.

All the properties of Radon measures hold. The theorem’s proof is valid in $\widehat{X}$, and the rest is almost trivial. There is a single exception: [cancelled:



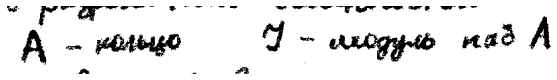
Original source details: GELFAND-P000015

Uncertain reading. св-ва ... бор. ... [properties ... Borel ...; visible fragmentary candidates beneath the horizontal stroke; the complete cancelled phrase is unreadable]

] spaces. — one cannot define the support of arbitrary Radon measures.

Full reading notes

GELFAND-0001-U001



Original source details: GELFAND-P000001

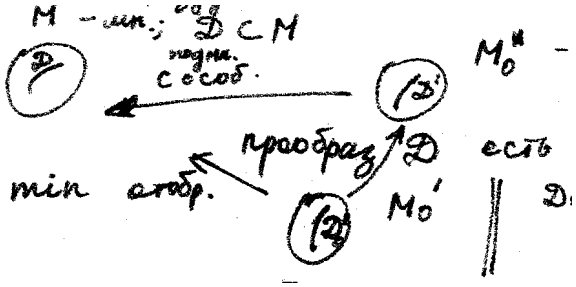
Observation. After “A is a ring” is a handwritten letter with a fork at the top and a lower hook; it recurs for the module and the ideal.

Candidates: $\mathcal{J}$; $\mathcal{I}$; J

Historical note: A calligraphic J was used in the text as a graphical candidate; the notation was not identified from its mathematical meaning.

Fresh source comparison: The same fork and lower hook are visible in three separate occurrences at 5×/10×. This supports a consistent graphical transcription but does not distinguish every possible conventional I/J letterform.

GELFAND-0001-U002



Original source details: GELFAND-P000001

Observation. In the upper half are three rounded contours with inner curves and two leftward arrows; D, D-prime, M and indices occur above and beside them.

Candidates: None proposed.

Historical note: Uncertain inner curves and joins were not redrawn. The description and the source-figure entry retain a locator for comparison.

Fresh source comparison: The complete diagram crop is now displayed at its typed description. Both leftward arrowheads are visible. A separate curved stroke rises from the lower contour toward the area beside D; its exact attachment to the upper contour is not asserted.

GELFAND-0001-U003



Original source details: GELFAND-P000001

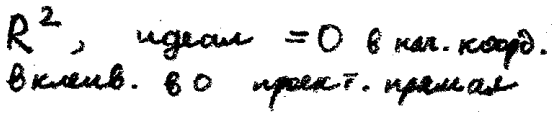
Observation. The upper index on M_0 resembles H; over M_0 at the lower region is a short stroke.

Candidates: M_0^H; M_0'; M_0^N; M_0'

Historical note: The indices and prime need comparison with the original.

Fresh source comparison: Separate close crops show the dark upper index and the lower prime at the reading itself. The lower prime is distinct; the dark upper sign still does not justify excluding N in favor of H.

GELFAND-0001-U006



Original source details: GELFAND-P000001

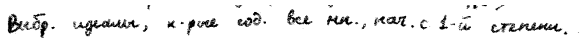
Observation. To the right of the R^2 example the line begins with a joined group of letters and ends “into 0, a projective line”.

Candidates: Вклеив. в 0 проект. прямой [Gluing a projective line into 0]; Оклеив. в 0 проект. прямой [alternative initial abbreviation]

Historical note: The phrase is given inside the uncertain-reading entry, without choosing from the mathematical meaning of a blow-up.

Fresh source comparison: The wider context and 5×/10× details strongly favor the initial В and the abbreviation Вклеив. The historical alternative remains visible; the source wording is not replaced by a standard construction of a blow-up.

GELFAND-0001-U007



Original source details: GELFAND-P000002

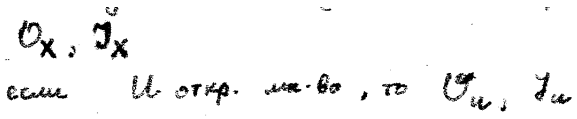
Observation. A rereading of the enlarged original fragment favors “polynomials, starting with degree 1”; the middle strokes of the abbreviation нач. are still not fully distinguishable.

Candidates: мн., нач. с 1-й степени [polynomials, starting with degree 1]; мн., к-ст. с 1-й степенью [alternative abbreviation; expansion uncertain]

Historical note: The first candidate was selected from the shape of the original line; it was not replaced by the standard definition of Proj.

Fresh source comparison: The line ending is visible in the wider crop and separate 5×/10× details. Its first abbreviation Выбр. is visible too, but its expansion could express selecting or discarding. The English text retains that abbreviation rather than deciding the operation from the construction. The strokes favor нач. at the end; the historical alternative remains recorded.

GELFAND-0001-U008



Original source details: GELFAND-P000002

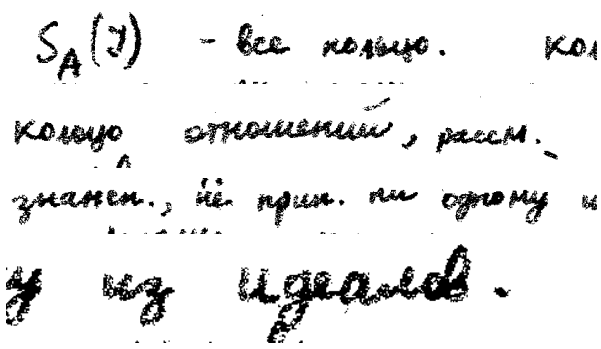
Observation. In a separate rereading, the first letter after “then” matches the shape of O_X above; the second matches the ideal letter with a lower hook.

Candidates: O_U, J_U; O_U, I_U

Historical note: The initial reading of the first letter as J was corrected from the scan. The possible I/J distinction in the second letter remains, independently of the standard sheaf construction.

Fresh source comparison: The corrected context now includes the whole lower subscripts. The rounded first sign and forked, hooked second sign match the earlier pair. These shapes support O followed by the ideal letter, without deciding every conventional I/J distinction. Separate 5×/10× details were inspected.

GELFAND-0001-U009



Original source details: GELFAND-P000002

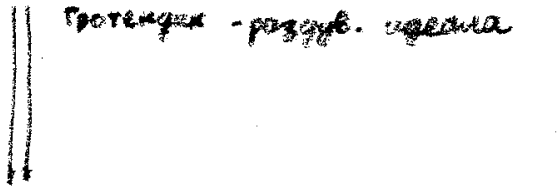
Observation. After “ring of fractions” one can see “numer., denom., not belonging to any of the ideals”, but the abbreviation boundaries and the scope of the last phrase are unclear.

Candidates: числ., знамен., не прин. ни одному из идеалов [numer., denom., not belonging to any of the ideals]; числ.; знамен. не прин. ни одному из идеалов [numer.; denom. not belonging to any of the ideals]

Historical note: The punctuation was not corrected from the definition of localization.

Fresh source comparison: Four overlapping source parts retain the line ending числ., the following знамен., and the phrase about the ideals. The abbreviations are visible at 5×/10×; their punctuation does not establish which nouns the restriction modifies. No missing qualifier is supplied.

GELFAND-0001-U010



Original source details: GELFAND-P000002

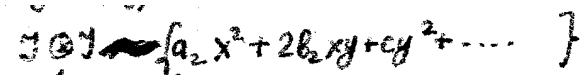
Observation. The right-hand note beside two vertical arrows reads “Grothendieck — blow up …”; the final word is written continuously.

Candidates: идеально [ideally]; идеалы [ideals]

Historical note: The final word is not asserted.

Fresh source comparison: The wider crop includes both downward arrows and the full note. The joined final strokes favor идеально, but do not securely exclude идеалы. The literal candidate “blow up ideally” is retained with the alternative; the wording is not regularized into a definition.

GELFAND-0001-U011



Original source details: GELFAND-P000002

Observation. After J tensor J is a thick two-sided mark and a quadratic polynomial; the coefficients resemble a_2, $2b_2$, c.

Candidates: $a_2x^2 + 2b_2xy + cy^2 + \cdots$; $a_2x^2 + 2b_2xy + c_2y^2 + \cdots$

Historical note: The index on the final coefficient and the direction of the thick mark are uncertain.

Fresh source comparison: The complete context and separate details show a_2 and $2b_2$. A small upper stroke is visible near the c-shaped coefficient; it is not a secure lower index 2. The thick connecting mark is retained as an uncertain graphical candidate, without inferring an isomorphism.

GELFAND-0001-U012



Original source details: GELFAND-P000002

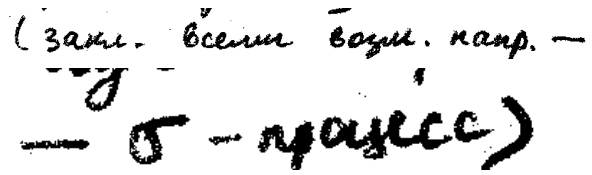
Observation. At the lower edge is $a_0 + (a_1x - b_1y)z + \cdots$; the horizontal sign before b_1y looks like a minus.

Candidates: $a_0 + (a_1x - b_1y)z + \cdots$; $a_0 + (a_1x + b_1y)z + \cdots$

Historical note: The visible minus is retained in the body; the next scan has a plus. The arrow at left is written heavily.

Fresh source comparison: The enlarged sign is a horizontal minus; no vertical crossing stroke is visible. The next source page has a plus. Both source readings remain, and the mathematical commentary does not use a change of sign to alter either transcription. The heavy arrow still has uncertain direction.

GELFAND-0001-U013



Original source details: GELFAND-P000003

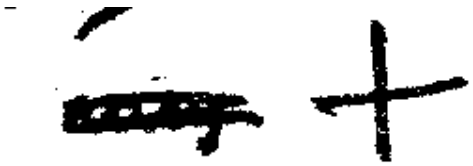
Observation. In parentheses after “plane” is the abbreviation “замк. всеми возм. напр.” and a sign resembling sigma before “process”.

Candidates: замк. всеми возм. напр. — σ-процесс [first abbreviation uncertain; with all possible directions — sigma-process]; зам. всеми возм. напр. — σ-процесс [alternative first abbreviation; with all possible directions — sigma-process]

Historical note: The abbreviations were not expanded or replaced by a term inferred from the meaning.

Fresh source comparison: The initial abbreviation and the sigma-like sign have been inspected at 5×/10×. The final short stroke of the first abbreviation is still not secure. The translation expands the readily legible neighboring abbreviations, but keeps замк. itself visibly unexpanded.

GELFAND-0001-U014



Original source details: GELFAND-P000003

Observation. Between “In the space” and the cross is a heavily cancelled short group of letters.

Candidates: None proposed.

Historical note: The cancellation itself is marked; the underlying text cannot be read confidently.

Fresh source comparison: The opaque horizontal cancellation remains unreadable even at 10×. The separate original cross is also shown beside the existing two-stroke native recreation. Its thick ends do not establish intentional arrowheads; the recreation retains two crossing, unarrowed strokes.

GELFAND-0001-U016



Observation. Under the name Dobrushin, before “Find”, a group of letters is completely crossed out.

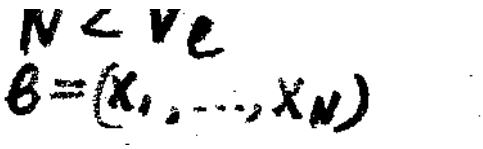
Original source details: GELFAND-P000003

Candidates: None proposed.

Historical note: The visible cancellation is preserved; the original word was not reconstructed.

Fresh source comparison: The wider context identifies the position before “Find”; the 5×/10× details show multiple dense crossing strokes. They do not support a complete word, so no candidate is invented.

GELFAND-0001-U017



Observation. Before $(x_1, \ldots, x_N)$, and later in the formulas, is a sign with a rounded lower part and a dark upper stroke.

Original source details: GELFAND-P000003

Candidates: θ; b; в

Historical note: The fragment uses theta as a single graphical candidate for this recurring sign.

Fresh source comparison: The corrected crop includes the entire left-hand sign and its equality. Two separate occurrences were inspected at 5×/10×. The divided loop and thick upper stroke are consistent, but do not securely exclude b or Cyrillic в. The theta transcription remains explicitly provisional.

GELFAND-0001-U018



Observation. Between “The Ising model” and $U(\theta)$ is a separate completely inked-out sign.

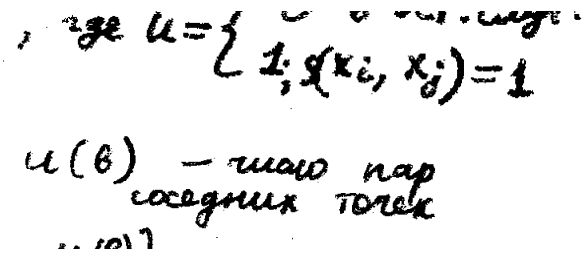
Original source details: GELFAND-P000003

Candidates: None proposed.

Historical note: The cancelled sign was not taken as a mathematical operator.

Fresh source comparison: The sign remains opaque at 5×/10×; its dark mass and the underlying strokes do not identify an operator or letter. Its location and cancellation are preserved, without assigning it a mathematical role.

GELFAND-0001-U019



Observation. In the second branch of u=..., the sign before $(x_i, x_j) = 1$ resembles rho; the caption below begins $u(\theta)$.

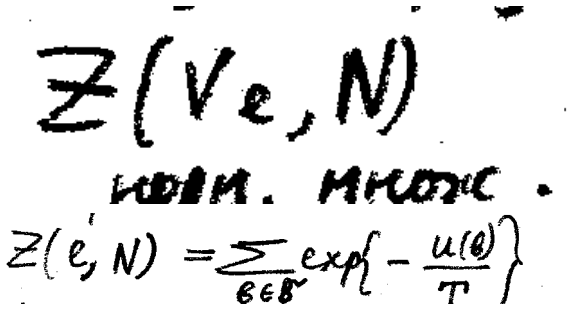
Original source details: GELFAND-P000003

Candidates: $\rho(x_i, x_j) = 1$; $\chi(x_i, x_j) = 1$

Historical note: A sign resembling a semicolon is visible after the unit value. The small u in the caption was not replaced with U.

Fresh source comparison: A corrected context contains the complete equality, and a repaired 5×/10× target shows the rho-shaped letter itself. The earlier narrow detail missed that target and is not used as evidence for it. The full lower caption clearly begins with a small u-shaped letter. The branch punctuation and the U/u difference remain diplomatic; their mathematical implications are discussed separately.

GELFAND-0001-U020



Observation. The first normalizing quantity looks like $Z(V_\ell, N)$; below is $Z(\ell, N)$. The set under the second sum has the shape of a handwritten B.

Original source details: GELFAND-P000003

Candidates: $Z(V_\ell, N)$; $Z(\ell, N)$; $\theta \in B$; $Z(V_\ell, N)$; $Z(\ell, N)$; $\theta \in \mathcal{B}$

Historical note: The difference in arguments is preserved. The set symbol was not identified with the event B from its meaning.

Fresh source comparison: Separate 5×/10× details confirm the visible V_ℓ in the first denominator and ell in the next line. The set under the second sum carries an added upper stroke, supporting a distinct graphical candidate, but ordinary B versus a calligraphic form remains uncertain. The transcription keeps the event and configuration-set candidates distinct.

GELFAND-0001-U021



Original source details: GELFAND-P000003

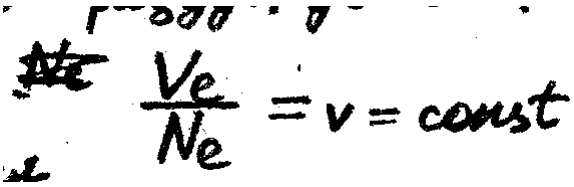
Observation. After “The molecules attract” are parentheses with inked signs, arrows, and the words “neighbors” and “probab.”.

Candidates: соседи > их вероятн. [neighbors > their probab.]; соседи > его вероятн. [neighbors > its probab.]

Historical note: The provisional caption was given only as a candidate; the drawing was not converted into a precise assertion.

Fresh source comparison: The filled marks, arrows and caption are displayed together. At 5×/10× the pronoun looks closer to ero than to их, but the elliptical phrase still does not support a complete assertion. The inherited body candidate remains visibly marked as uncertain and both historical alternatives are retained.

GELFAND-0001-U022



Original source details: GELFAND-P000003

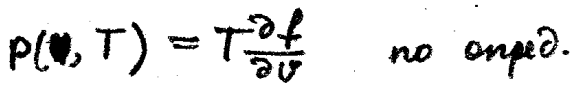
Observation. Before V_ℓ/N_ℓ is a heavily cancelled short mark.

Candidates: None proposed.

Historical note: The original sign was not recovered.

Fresh source comparison: The cancelled mark remains unreadable in the enlarged details. The wider source fragment includes the fraction and $v = \text{const}$, so its place is inspectable. The written ratio V_ℓ/N_ℓ is retained, rather than silently changed to particle density N_ℓ/V_ℓ.

GELFAND-0001-U024



Original source details: GELFAND-P000004

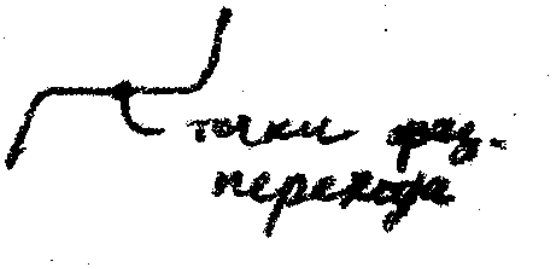
Observation. The first argument of $p(\cdot, T)$ is heavily written. The fraction contains $\partial f/\partial v$ and is preceded by T; no minus is visible before the derivative.

Candidates: $p(v,T) = T\partial f/\partial v$; $p(\nu,T) = T\partial f/\partial v$

Historical note: No thermodynamically expected correction of the sign was substituted.

Fresh source comparison: The whole frame and 3×/5×/10× details show no minus before T. The first argument remains a heavily written v-like sign; ν is not excluded. A separate commentary derives a possible plus under an explicit definition of f; this does not settle the letter or amend the source.

GELFAND-0001-U025



Original source details: GELFAND-P000004

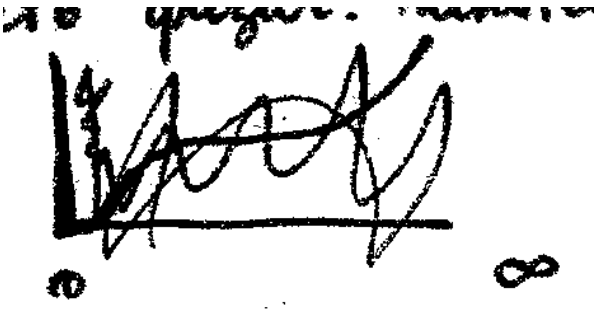
Observation. The upper right drawing has a rising curve with a flat middle, an arrow toward it, and the caption “points of phase transition” [точки фаз. перехода].

Candidates: None proposed.

Historical note: The precise axes, points, and shape of the transitions are not identified confidently.

Fresh source comparison: After extending the crop downward, the whole caption “точки фаз. перехода” is visible, in the plural. No axes are visible. The typeset sketch represents the flat middle, point, and curved pointer; it does not assert the precise direction of the heavily inked arrowhead.

GELFAND-0001-U026



Original source details: GELFAND-P000004

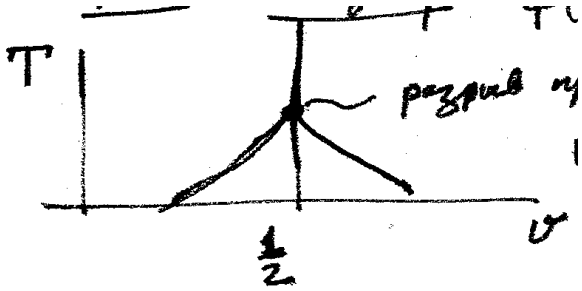
Observation. At left are two axes, with 0 and ∞ by the horizontal axis; several heavily overwritten curves and straight lines, and a vertical letter.

Candidates: vertical label φ; vertical label f

Historical note: The crossed-out lines were retained by description; no precise underlying curve was selected.

Fresh source comparison: Even at 10×, individual connections of the underlying curves cannot be separated from the heavy cancellations. The partial typeset sketch shows only the secure axes, 0 and ∞; the disputed vertical letter remains among the candidates. The neighboring clean graph at right is a different drawing and is shown separately.

GELFAND-0001-U028



Original source details: GELFAND-P000004

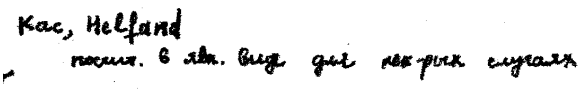
Observation. Axes T and v; a curve with a vertex, a vertical through the vertex, $1/2$ below, and a small wavy branch at right.

Candidates: None proposed.

Historical note: The branches and their connections require the original drawing; an exact redrawing had not been made.

Fresh source comparison: The left branch has an additional stroke; the vertical passes through the vertex, with $1/2$ below. A small visible gap separates the short wavy line at right from the vertex. The typeset sketch retains that gap without assigning physical meaning to the line.

GELFAND-0001-U029



Original source details: GELFAND-P000004

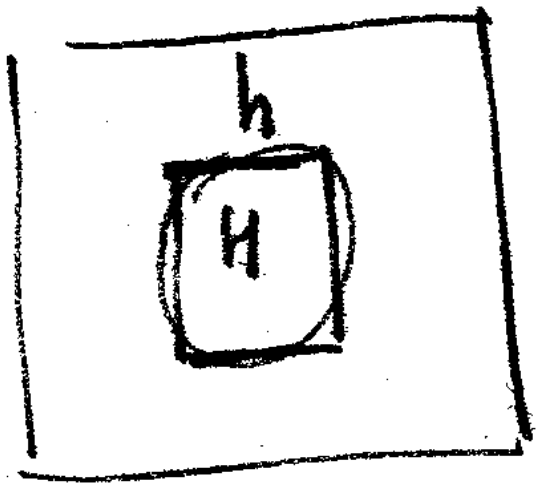
Observation. A repeated twofold enlargement of the line beneath Kac, Helfand shows “точн. в яв. виду для нек-рых случаях” [exact, in explicit form, for some cases]; the word endings are not completely clear.

Candidates: точн. в яв. виду для нек-рых случаях [literal source endings]; точн. в яв. виде для нек-рых случаев [alternative endings]

Historical note: The word случая... [case...] has now been read, replacing the erroneous initial сущ.; neither grammar nor the known result is used to correct the ending silently.

Fresh source comparison: The wide crop retains the names and the entire line. The 5×/10× details favor виду and случаях. The historical alternatives виде and случаев remain in full; the English body conveys the meaning while separately displaying the nonstandard source endings.

GELFAND-0001-U030



Observation. A large square and a smaller outlined region, H at the center and h above; the small contour has overlapping rounded and straight strokes.

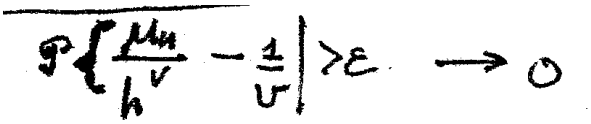
Original source details: GELFAND-P000004

Candidates: None proposed.

Historical note: The source was retained as a drawing, without selecting an exact boundary from the term “subcube”.

Fresh source comparison: Both straight and rounded strokes of the inner boundary are visible at 5×/10×; h above H is readable. The outer lines have gaps. The typeset sketch retains the overlap and gaps rather than replacing the drawing with a perfect square or circle.

GELFAND-0001-U031



Observation. A vertical bar is visible before $> \varepsilon$; before the first fraction, a possible left bar merges with the brace.

Original source details: GELFAND-P000004

Candidates: $\mathsf{P}\{\mu_H/h^\nu - 1/v \mid > \varepsilon\} \to 0$; $\mathsf{P}\{|\mu_H/h^\nu - 1/v| > \varepsilon\} \to 0$

Historical note: The matching bar was not added silently; the visible candidate was left in the body.

Fresh source comparison: The right vertical stroke is separate from $> \varepsilon$. An independent left absolute-value bar beside the opening brace cannot be established even at 10×. The next frame clearly has paired bars, but that comparison does not change this frame’s notation.

GELFAND-0001-U032

Observation. At the lower right edge, a small isolated sign resembles 2 or a fragment of a letter.



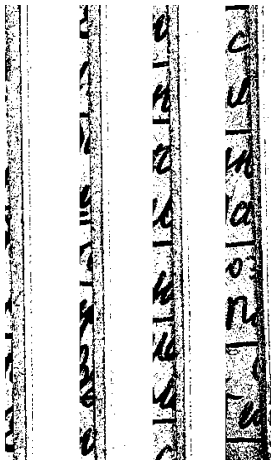
Original source details: GELFAND-P000004

Candidates: 2; fragment of a letter

Historical note: No confident page number was assigned.

Fresh source comparison: At 5×/10×, only a short hook and fragmentary edge strokes remain. The digit 2 and a fragment of a letter remain candidates; no number is assigned.

GELFAND-0001-U033



Observation. At the extreme left edge of the frame, a narrow strip of neighboring text is cut off to isolated endings and strokes; a separate crop at native pixel resolution was checked.

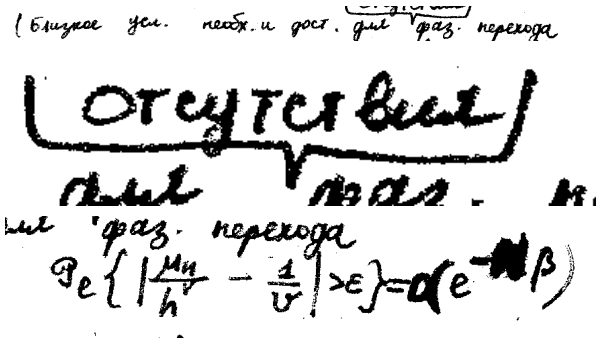
Original source details: GELFAND-P000004

Candidates: None proposed.

Historical note: Connected text is unavailable in this frame; the page has unreadable status because of this portion. The main writing surface has been read and transcribed.

Fresh source comparison: Four overlapping source strips retain the entire available left edge. It is already clipped in the whole original frame; further magnification cannot restore the missing portions of letters. No independently readable words were established.

GELFAND-0001-U034



Observation. In $O(e^{\cdots\beta})$, the sign between the minus and β is heavily obscured; before the preceding phrase is an opening round parenthesis without a clearly visible closing parenthesis.

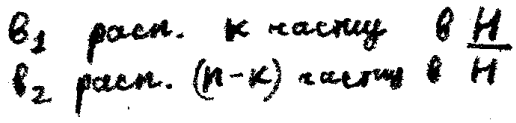
Original source details: GELFAND-P000005

Candidates: $O(e^{-N\beta})$; $O(e^{-\ell\beta})$

Historical note: The obscured sign was not declared settled, and a missing parenthesis was not added to the source text.

Fresh source comparison: The corrected crop includes the tops of all overlapping strokes in the exponent. The minus and β are visible; the intervening sign cannot reliably be distinguished as N or ℓ at 10×. Separate crops show the insertion “absence” and the opening parenthesis. The insertion is retained explicitly, and the parenthesis is not closed editorially.

GELFAND-0001-U035



Observation. In the line about θ_2 and $N-k$ particles, a short upper stroke is visible by H.

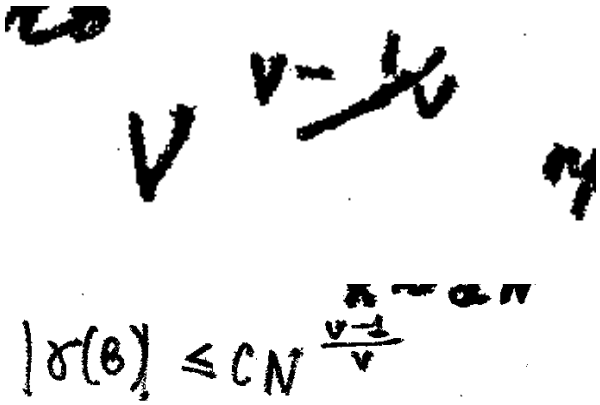
Original source details: GELFAND-P000005

Candidates: $\overline{H}$; H

Historical note: The barred candidate was used in the text; the complement was not determined solely from its meaning.

Fresh source comparison: The wider crop shows both H labels and the horizontal stroke between them. It may be an overbar on the lower H or an underline on the upper H; the 5×/10× details do not settle its attachment. The historical barred candidate remains a candidate, not an established reading.

GELFAND-0001-U036



Observation. After the words about the number of boundary points, the exponent of V is tightly written: ν-1, a slanted fraction stroke, and ν.

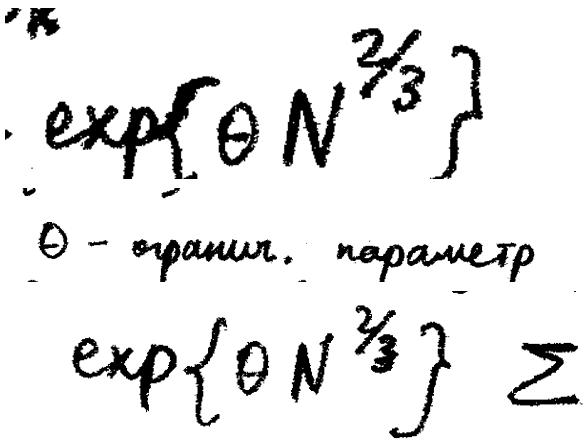
Original source details: GELFAND-P000005

Candidates: $V^{\nu-1/\nu}$; $V^{(\nu-1)/\nu}$

Historical note: The first candidate was retained. In the following bound involving N, the exponent is written more clearly as a fraction.

Fresh source comparison: The two crops show the tight exponent on V and the explicit common fraction $(\nu-1)/\nu$ on N. Their difference is retained. A separate mathematical commentary derives boundary scaling for an explicitly defined lattice model; this derivation is not substituted for reading the ink.

GELFAND-0001-U037



Observation. In the factor $\exp\{\cdots N^{2/3}\}$, the letter resembles θ; below it, the same letter is explained as “bounded parameter” [огранич. параметр].

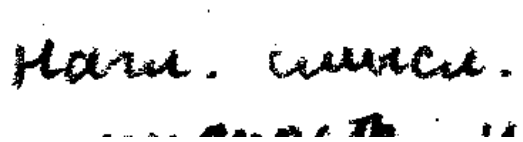
Original source details: GELFAND-P000005

Candidates: θ; Θ

Historical note: It may differ from the configuration symbol; their identity has not been declared proved.

Fresh source comparison: Both factors and the caption show an oval sign with a transverse stroke. The θ/Θ case and its distinction from the configuration symbol remain unsettled. The exponent $2/3$ is clearly visible; it is not replaced by the general exponent written above.

GELFAND-0001-U073



Original source details: GELFAND-P000005

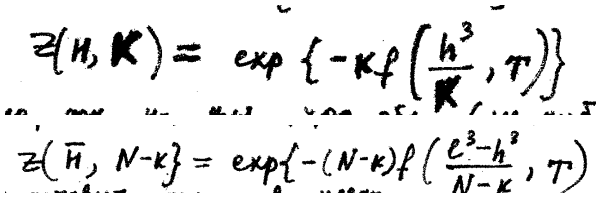
Observation. The first explanatory line after the phase-transition condition begins with an abbreviation best read as “Нагл. смысл.” [Нагл. meaning]; its middle strokes merge.

Candidates: Нагл. смысл. [Нагл. meaning]; Нач. смысл. [Нач. meaning]

Historical note: The abbreviation was not expanded into a complete word; the candidate was selected from the original fragment at 3× magnification.

Fresh source comparison: The whole word and 5×/10× details favor Нагл. The alternative Нач. and the historical uncertainty about the middle strokes remain recorded; no full word is supplied.

GELFAND-0001-U039



Original source details: GELFAND-P000006

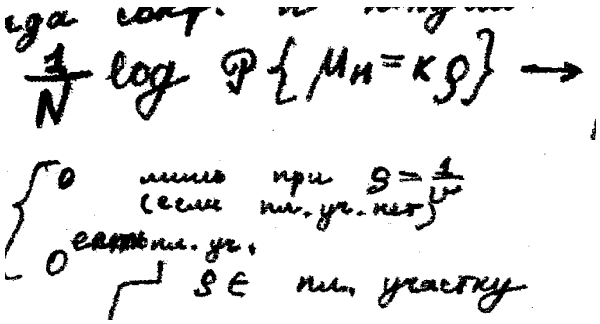
Observation. In the argument of f, a visible exponent 3 stands above h; the following line also has ℓ^3-h^3.

Candidates: $h^3/k;\ (\ell^3 - h^3)/(N-k);\ h^{\nu}/k;\ (\ell^{\nu} - h^{\nu})/(N-k)$

Historical note: The visible 3 was retained, without standardizing it to the dimension symbol ν on neighboring lines.

Fresh source comparison: The whole frame, both contexts and 5×/10× details support 3 on h and ℓ. The readings h^3 and ℓ^3-h^3 have graphical support; neighboring ν does not justify replacing them. The historical alternatives remain recorded.

GELFAND-0001-U040



Original source details: GELFAND-P000006

Observation. The probability argument has μ_H=k followed by a rounded sign; at right are a density sign and two branches with zeros. Text above the second branch is inked over.

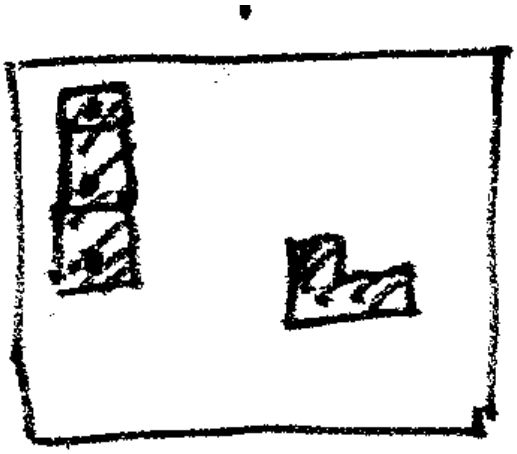
Candidates: $\mu_H = k\rho;\ \rho = 1/v;\ \mu_H = k\theta;\ \rho = 1/v;\ \mu_H = k\delta;\ \delta = 1/v$

Historical note: The inked-over text is unreadable. The two visible zeros were not replaced by different limits according to an expected meaning.

Fresh source comparison: The rounded letter after k has a curved descending stem, supporting ρ; θ and δ remain historical alternatives. Both zeros are retained. The

inked-over writing above the second branch remains unreadable at 10×. The small step-like drawing is shown separately.

GELFAND-0001-U041



Observation. Inside a large square frame, a vertical chain of shaded cells is at left, and an angular group of cells at right.

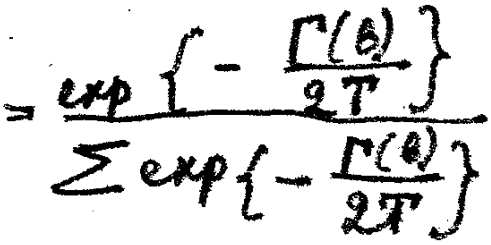
Original source details: GELFAND-P000006

Candidates: None proposed.

Historical note: The exact cell count and internal marks are uncertain; the original drawing is required.

Fresh source comparison: Comparing the rendered page with the 10× details does not securely distinguish the left chain's internal divisions from its dense hatching. The provisional new count of three cells was overconfident and was withdrawn before publication. At right, three cells form a corner. The partial typeset sketch at left shows only its outer outline and hatching; the cell count remains unresolved. The original remains alongside.

GELFAND-0001-U042



Observation. In the two exponentials after the equality, a Γ-like sign has a possible prime at its upper right.

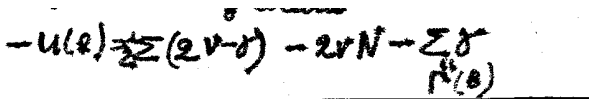
Original source details: GELFAND-P000006

Candidates: $\Gamma'(\theta)$; $\Gamma(\theta)$

Historical note: The primed candidate was used in the body despite the visible caption referring to length/area.

Fresh source comparison: The corrected r3 details favor Γ without a separate prime in both places; strokes overlap near the lower parenthesis. The initial r2 details clipped the tops and do not decide the prime. Both alternatives and the historical body candidate remain; the definition of length does not substitute for reading the ink.

GELFAND-0001-U043



Original source details: GELFAND-P000006

Observation. The bottom formula begins -U(θ), followed by an overwritten sign before $\Sigma(2\nu\text{-}\delta)$, then 2νN and a sum of δ with lower index $\Gamma'(\theta)$.

Candidates: $-U(\theta) \geq \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta$

$-U(\theta) \leq \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta$

$-U(\theta) = \sum(2\nu - \delta) = 2\nu N - \sum_{\Gamma'(\theta)} \delta$

Historical note: The energy normalization was not corrected to the standard model; the index and final sign also require checking.

Fresh source comparison: At 10× the crossed strokes before the sum do not uniquely determine $\geq$, $\leq$ or =. The last sum's lower index is heavily written over. A separate edge count checks a possible factor of two under explicit conventions and does not amend this line.

GELFAND-0001-U044



Original source details: GELFAND-P000007

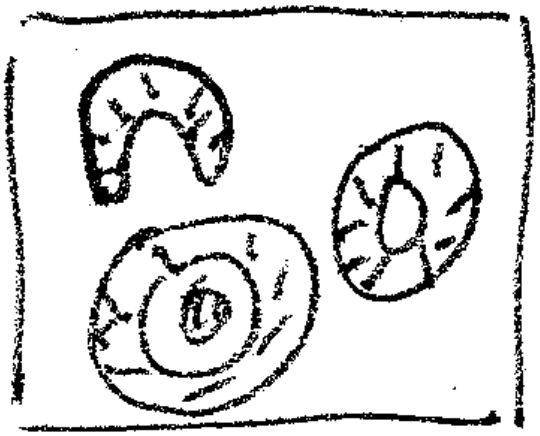
Observation. A small square frame at right contains two densely filled rounded patches and several strokes in its upper left corner.

Candidates: None proposed.

Historical note: The small strokes and inked-in internal contours cannot be enumerated confidently.

Fresh source comparison: The two large patches and their white openings are distinguishable. Several short strokes at upper left stand above a curved lower stroke; they are not assigned a definite particle count or letter reading. A partial typeset recreation preserves the arrangement and the uncertainty of those marks.

GELFAND-0001-U045



Original source details: GELFAND-P000007

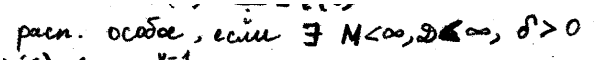
Observation. A large frame at left contains three separate rounded groups with internal contours and hatching; one contains a central island.

Candidates: None proposed.

Historical note: The nested cycles and hatching were not replaced by an approximately invented figure.

Fresh source comparison: The full context and new details distinguish an upper-left horseshoe with a white gap open below, a right-hand ring, and a lower-left group with a central island. The typeset sketch preserves that nesting and open gap. It does not claim to recover every pen irregularity or hatch endpoint.

GELFAND-00001-U046



Original source details: GELFAND-P000007

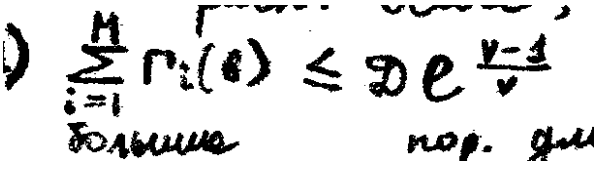
Observation. After ∃, the first finite parameter resembles N; the summation limit in condition 1 resembles M.

Candidates: $\exists N < \infty,\ D < \infty,\ \delta > 0;\ \sum^{M}$; $\exists M < \infty,\ D < \infty,\ \delta > 0;\ \sum^{M}$

Historical note: The difference between the letters was not removed on the basis of the repeated definition in later scans.

Fresh source comparison: In the full context the first letter still favors N, while the later limit favors M. The small r2 detail shows only $<\infty$ and cannot by itself decide the letter. The difference is not reconciled with the subsequent definition.

GELFAND-0001-U047



Original source details: GELFAND-P000007

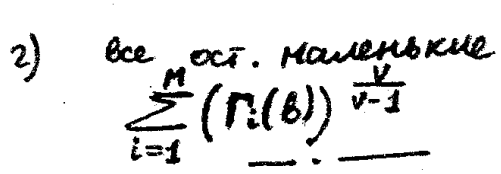
Observation. The exponent on the length in the first condition has a small fraction bar: ν-1 above ν.

Candidates: $\ell^{(\nu-1)/\nu}$; $\ell^{\nu-1}$

Historical note: The fractional candidate was retained. GELFAND-P000010 visibly has exponent ν-1; it was not imported here.

Fresh source comparison: The 5×/10× details clearly show the fraction bar and a separate ν below it, supporting (ν-1)/ν. The historical alternative and the difference from frame 10 remain recorded; the expected surface exponent does not alter the ink.

GELFAND-0001-U048



Original source details: GELFAND-P000007

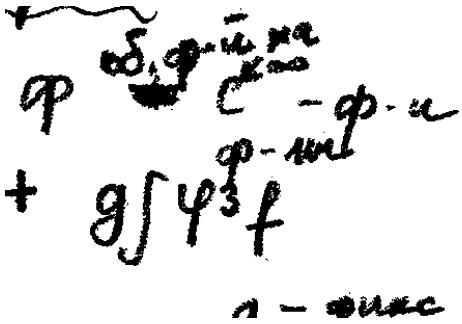
Observation. The second sum visibly has upper M and lower i=1; its caption says “all the others small”, and no continuation follows the power.

Candidates: $\sum_{i=1}^{M} \Gamma_i^{\nu/(\nu-1)}$; $\sum_{i=M+1}^{\infty} \Gamma_i^{\nu/(\nu-1)}$

Historical note: Only the visible limits were put into the body; no condition was added from a later scan.

Fresh source comparison: The context preserves the upper letter and caption; 5×/10× details confirm lower i=1. No continuation follows the expression before the dividing line. The sum starting at i=M+1 and the bound in frame 10 are not substituted.

GELFAND-0001-U050



Original source details: GELFAND-P000007

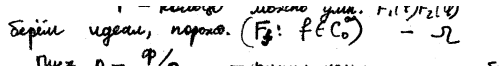
Observation. At right of the equation are several lines and an arrow near φ and C_0^∞; the upper part resembles “б, ф-й на”.

Candidates: б, ф-й на C_0^∞ [fragment: functions on C_0^∞]; об. ф-й на C_0^∞ [source abbreviation: functions on C_0^∞]

Historical note: The text was not expanded into a definition of a distribution on semantic grounds.

Fresh source comparison: The enlarged initial sign admits “об.”, but overlapping strokes do not securely expand the abbreviation. C_0^∞ and the arrow are distinguishable. The separate “φ — мн.” also remains abbreviated. No definition of a generalized function is inserted from context.

GELFAND-0001-U051



Original source details: GELFAND-P000007

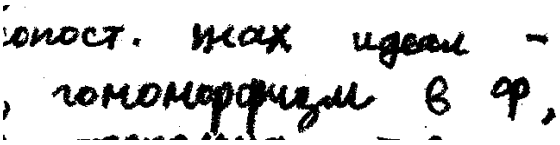
Observation. In “Take the ideal, generated by (F_: f∈C_0^∞)”, the sign after the dash resembles Ω; the index on F is densely written.

Candidates: F_f; Ω; F_j; Ω; F_f; J

Historical note: The symbols remain candidates and were not corrected through the algebraic construction.

Fresh source comparison: The corrected context extends rightward through Ω. The sign after the dash resembles a broad Ω with lower ends. The F index merges with neighboring strokes, so f remains a preference rather than an established reading. Algebra does not decide the letters.

GELFAND-0001-U052



Original source details: GELFAND-P000007

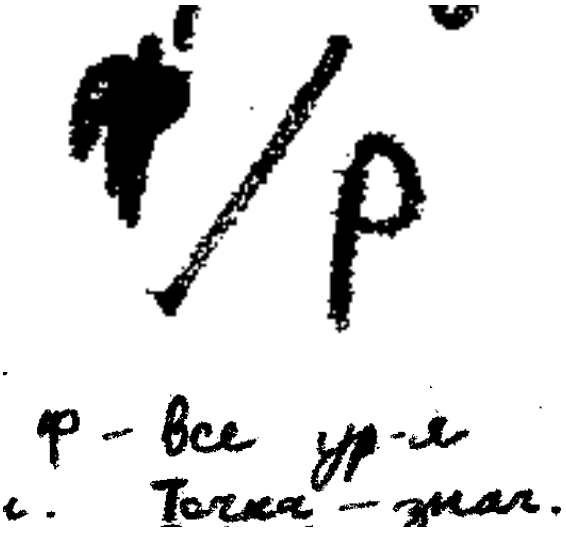
Observation. After “a homomorphism into”, a rounded letter with a vertical stroke resembles Φ.

Candidates: Φ; a field denoted by a similar-looking letter

Historical note: It was not automatically replaced by a number field.

Fresh source comparison: At 5×/10× two rounded parts and a vertical stem support Φ graphically. A field is not supplied in the diplomatic line. The separate evaluation-map commentary states its own assumptions and does not correct the source notation.

GELFAND-0001-U053



Original source details: GELFAND-P000008

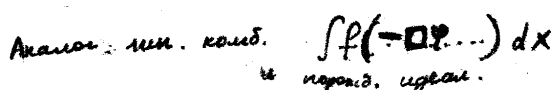
Observation. Above p in the fraction is a dark letter with an upper horizontal and a rounded part; the similar letter in the caption at right is followed by “all equations”.

Candidates: Φ; P; A

Historical note: The text used Φ as a candidate, without substituting a conventional letter for a polynomial ring.

Fresh source comparison: Two separate crops show the numerator and caption without shrinking the entire line. The caption has a rounded part and stem, while the numerator is heavily overwritten. Φ remains a candidate; a conventional polynomial-ring symbol is not reconstructed from theory.

GELFAND-0001-U054



Original source details: GELFAND-P000008

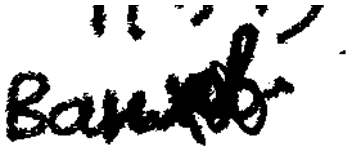
Observation. The integral is f(-□...)dx; the sign immediately after the box is heavily overlapped and resembles φ, followed by dots.

Candidates: $\int f(-\Box\varphi \ldots)\, dx$; $\int f(-\Box \ldots)\, dx$

Historical note: No omitted operator or additional term was invented.

Fresh source comparison: The full context retains the dense sign immediately after □; the following dots are separate ink marks at 5×/10×. They do not recover an additional equation term. Both historical readings are given alongside the integral.

GELFAND-0001-U055



Original source details: GELFAND-P000008

Observation. Beneath the limit, the surname begins “Ван”; the remainder is densely overwritten.

Candidates: Ван Хов [Van Hove]; Ванхов [Vankhov; joined source spelling]

Historical note: The space and final letters require checking.

Fresh source comparison: The beginning “Ван” is readable in context; the enlarged remainder has overlapping loops and does not settle the original spacing or ending. The English name Van Hove renders the candidate; it is not new graphical evidence.

GELFAND-0001-U056



Original source details: GELFAND-P000008

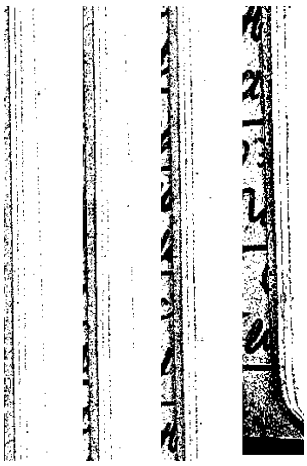
Observation. At lower right, between T and const, the sign consists of two horizontals with a possible diagonal stroke.

Candidates: $T = \text{const}$; $T \neq \text{const}$

Historical note: The equality candidate was not declared certain on the basis of the mathematical context.

Fresh source comparison: The correctly targeted r3 details show two horizontals crossed by a long descender from “окр.” above. Equality therefore has better graphical support, with the overlap retained as a limitation. The r2 details of neighboring words are not evidence about the sign.

GELFAND-0001-U057



Observation. At the extreme left edge of the frame is a cropped column of strokes and endings from neighboring handwriting; no complete words are visible.

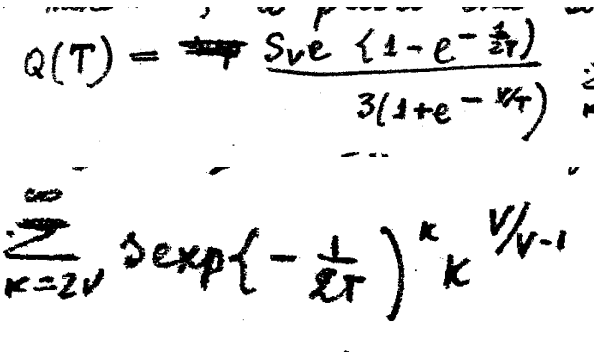
Original source details: GELFAND-P000008

Candidates: None proposed.

Historical note: This part cannot be transcribed from the available frame. The main page was read; “unreadable” applies to the cropped edge strip.

Fresh source comparison: Four overlapping strips preserve the available frame edge. The lower-strip details at 5×/10× show isolated endings and border noise rather than complete words. No data exist beyond the source raster; missing text is not invented.

GELFAND-0001-U059



Observation. After Q(T)= is a cancelled sign. A fraction with S_ν e is followed by a sum from k=2ν, then 3 exp{-1/(2T)} with superscript k and another factor k^{ν/(ν-1)}.

Original source details: GELFAND-P000009

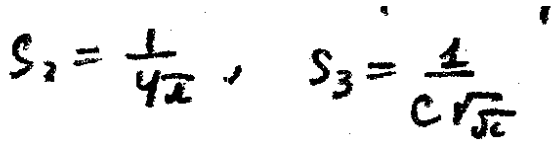
Candidates: $\frac{S_\nu e(1-e^{-1/(2T)})}{3(1+e^{-\nu/T})} \sum_{k=2\nu}^{\infty} 3\exp\{-1/(2T)\}^k k^{\nu/(\nu-1)}$

$\frac{S_\nu e(1-e^{-1/(2T)})}{3(1+e^{-\nu/T})} \sum_{k=2\nu}^{\infty} (3\exp\{-1/(2T)\})^k k^{\nu/(\nu-1)}$

Historical note: The summation limit, scope of exponent k and cancelled fragment require checking. An expected convergent formula was not substituted.

Fresh source comparison: The formula is split into two overlapping crops. Details support a separate superscript k by the closing brace and an ∞-like upper limit. The cancelled fragment remains unreadable. The two scopes of k have different mathematical consequences, discussed separately without choosing the ink reading by convergence.

GELFAND-0001-U060



Original source details: GELFAND-P000009

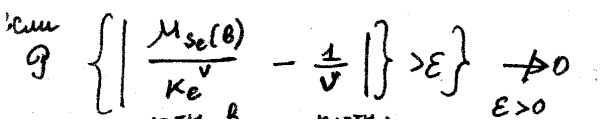
Observation. In the denominator of S_3 before $\sqrt{\pi}$, the sign resembles C, c or the digit 6.

Candidates: $1/(C\sqrt{\pi})$; $1/(c\sqrt{\pi})$; $1/(6\sqrt{\pi})$

Historical note: The number was not reconstructed from the known volume of a ball.

Fresh source comparison: At 5×/10× the sign is open to the right and looks more like C/c than a closed 6; size does not settle capitalization. All three historical alternatives remain. The separately computed value for a ball of surface area one is not presented as an ink reading.

GELFAND-0001-U062



Original source details: GELFAND-P000009

Observation. After the absolute value of the fraction minus 1/v is a closing brace before $>\varepsilon$ and another closing brace after ε, followed by a slashed arrow to 0.

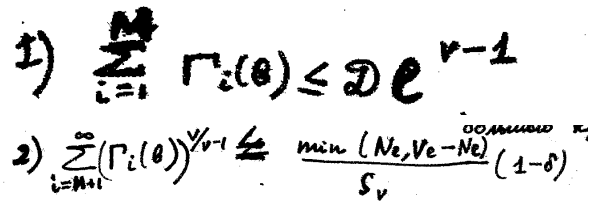
Candidates: $\mathrm{P}\{|\mu/K_\ell^\nu - 1/v|\} > \varepsilon\} \not\to 0$

$\mathrm{P}\{|\mu/K_\ell^\nu - 1/v| > \varepsilon\} \not\to 0$

Historical note: Both visible closing braces were retained; the expression was not corrected to the usual definition of a probability event.

Fresh source comparison: Both closing braces are distinguishable at 5×/10×. The early brace is not removed to make a well-formed probability event; the usual notation is only an alternative interpretation. The slashed arrow is also retained.

GELFAND-0001-U063



Original source details: GELFAND-P000010

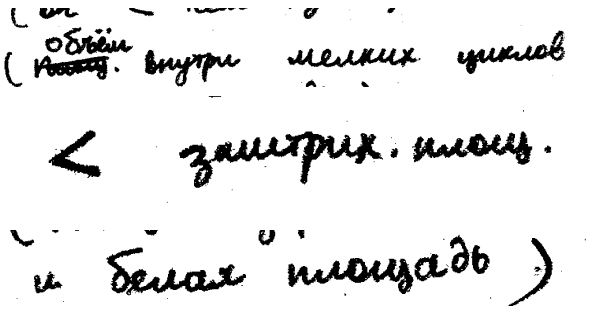
Observation. In condition 2 the sign before the fraction is heavily overwritten; the upper limit in condition 1 is a letter with superimposed strokes.

Candidates: $\leq;\ M;\ <;\ M;\ \leq;\ N$

Historical note: The indices and strictness of the inequality require comparison; the first candidate was retained in the body.

Fresh source comparison: Separate crops show both conditions, including the upper limit and denominator S_ν. A lower horizontal beneath the second relation supports $\leq$, while the overwriting remains. The upper letter is not uniquely established; neighboring M is not treated as proof.

GELFAND-0001-U064



Original source details: GELFAND-P000010

Observation. In the parenthesis before “inside small cycles”, a word is cancelled, with “volume” inserted above; the following lines use “area” in abbreviated and full forms.

Candidates: None proposed.

Historical note: The original cancelled word is unreadable. “Area” was not replaced with “volume” on dimensional grounds.

Fresh source comparison: Three overlapping crops preserve the inserted “volume”, the cancelled word, “shaded area” and “white area”. Loops and connections remain beneath the cancellation, but they do not establish a complete word at 10×. Area and volume are not standardized.

GELFAND-0001-U065



Original source details: GELFAND-P000010

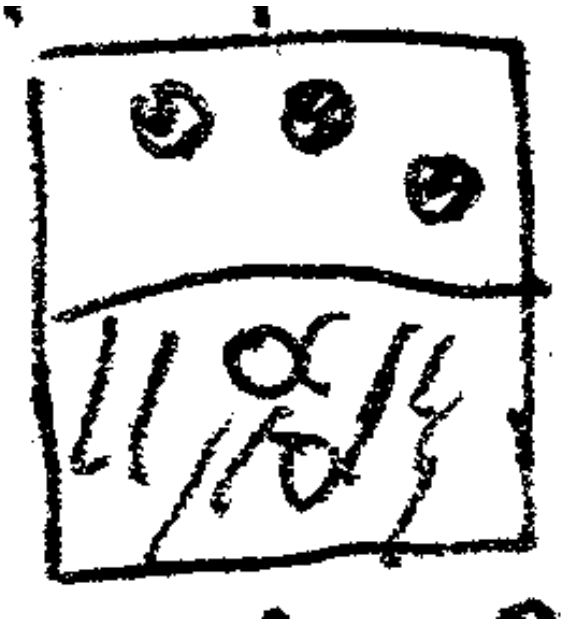
Observation. Rounded nested contours have a densely shaded outer ring and separate inner strokes.

Candidates: None proposed.

Historical note: The exact number of contours and the shaded and white components require the original.

Fresh source comparison: At 5×/10× four nested boundaries are distinguishable: two around the outer ring and two around the inner ring. The earlier editorial description of three contours is refined in a separate change journal. The recreation preserves four boundaries and two rings; individual merged pen strokes remain visible in the original.

GELFAND-0001-U066



Observation. A curve divides a rectangle: three small rounded regions are above; hatching and rounded white regions are below.

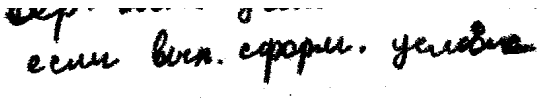
Original source details: GELFAND-P000010

Candidates: None proposed.

Historical note: The exact number of lower regions and the boundary's shape are uncertain.

Fresh source comparison: The full crop shows three upper patches with white openings. Below are one closed white hole and another partially outlined region among the strokes. The typeset sketch follows the visible lines without closing the uncertain lower region or claiming each boundary's exact shape.

GELFAND-0001-U068



Observation. After "if ... holds", the handwritten abbreviation before "condition" resembles "сформ." or "строгое".

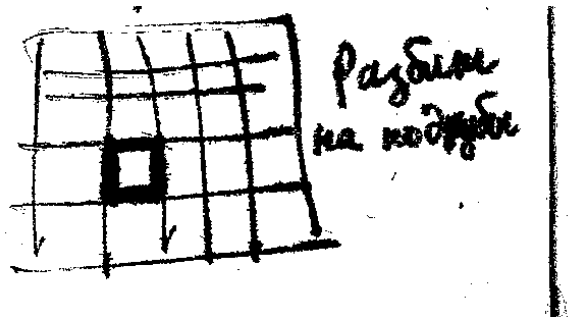
Original source details: GELFAND-P000010

Candidates: если вып. сформ. условие [if the stated condition holds]; если вып. строгое условие [if the strict condition holds]

Historical note: The sufficient condition's content was not reconstructed from the mathematics.

Fresh source comparison: The full context and 5×/10× details favor "сформ.": a rounded beginning and two descending middle strokes are distinguishable. "строгое" remains a historical alternative. No missing statement of the sufficient condition is supplied.

GELFAND-0001-U069



Observation. At right is a rectangular grid with a thick square in a central cell and caption “Divided into subcubes”.

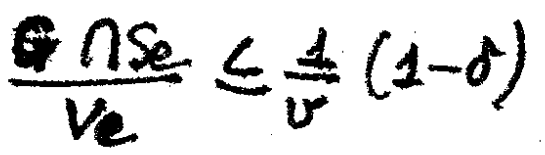
Original source details: GELFAND-P000010

Candidates: None proposed.

Historical note: The number of cells and exact highlighting are uncertain; the description does not specify invented coordinates.

Fresh source comparison: The corrected crop includes the whole caption. Six vertical and six horizontal lines give a five-by-five grid; the second cell from the left in the fourth row from the top is emphasized. “Central” was an editorial description, not a manuscript word; it is refined with a reversible journal. The sketch’s cell scale is schematic.

GELFAND-0001-U070



Observation. At the bottom edge is a formula with a dense sign before ∩S_ℓ, denominator V_ℓ and ≤1/v(1-δ).

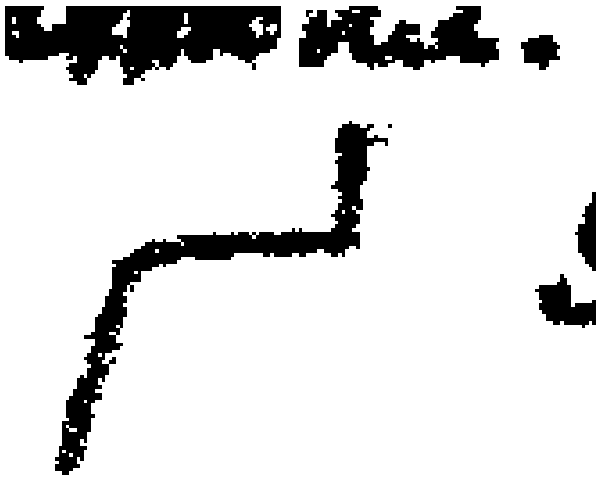
Original source details: GELFAND-P000010

Candidates: $(G \cap S_\ell)/V_\ell \leq (1/v)(1-\delta)$; $\mathrm{vol}(G \cap S_\ell)/V_\ell \leq (1/v)(1-\delta)$

Historical note: Neither a measure of the intersection nor the subcube’s volume was inserted automatically; the visible denominator was retained.

Fresh source comparison: The crop extends downward, and the new r3 detail preserves the whole V_ℓ. The numerator remains a heavily written G-like sign; no measure notation is visible. The subcube-averaging consequence is discussed separately and is not used to replace the denominator or insert a measure.

GELFAND-0001-U071



Original source details: GELFAND-P000006

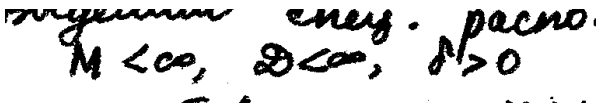
Observation. At right of the second limit branch is a small rising stroke, a horizontal segment to the right and another rise at the right end.

Candidates: None proposed.

Historical note: Separate axes, arrows and labels have not been established. The sketch was given as a source description without substituting a standard function or mathematical value.

Fresh source comparison: The 5×/10× details confirm a rising line, a nearly horizontal segment and a rise at right. No separate axes or labels are visible. The typeset stroke follows only the visible shape, with no function name or assigned values.

GELFAND-0001-U072



Original source details: GELFAND-P000009

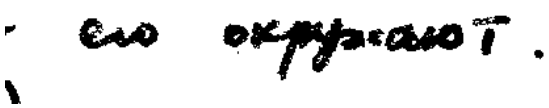
Observation. In the line of three numbers, the first letter has two tall vertical strokes and a thick inner connection. After reinspection at 2× and 10× it is closer to N, but M is not excluded.

Candidates: $N < \infty,\ D < \infty,\ \delta > 0$; $M < \infty,\ D < \infty,\ \delta > 0$

Historical note: The letter was not determined from an expected number of contours, nor automatically reconciled with M in the next frame's sum.

Fresh source comparison: The new narrow context and 5×/10× details support an N-like form, but the dense joining stroke leaves M as an alternative. The next frame's summation limit does not change this line. The historical report of earlier magnifications remains historical rather than being attributed to this new inspection.

GELFAND-0002-U001



Original source details: GELFAND-P000011

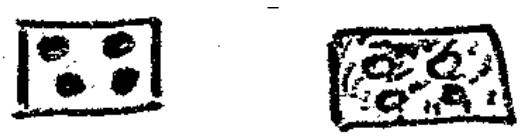
Observation. Before the verb "surround" is a short two-loop sign resembling co or ∞.

Candidates: co surround; ∞ surround

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (820, 20, 1270, 105). The contour context does not determine the two-loop sign or an omitted object of the surrounding; both graphical candidates are retained.

Fresh source comparison: The whole frame and 5×/10× details show a left open loop joined to a right loop. They do not uniquely distinguish the source syllable co from an infinity sign. No object of “surround” is supplied.

GELFAND-0002-U002



Original source details: GELFAND-P000011

Observation. The left rectangle contains four dark circles. In the right are four round marks: three with clear light centres; the lower right is darker. Additional curved strokes lie between them.

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (450, 75, 870, 175). A contour or probabilistic interpretation of the drawings is possible, but does not authorize changing the circles or inventing directions for the strokes.

Fresh source comparison: Four main marks in each rectangle are distinguishable at 10×. The left marks have small white pen gaps; the right lower mark has a smaller opening and surrounding strokes. A partial native sketch preserves the four-per-box arrangement; the small interior strokes remain in the original.

GELFAND-0002-U003



Original source details: GELFAND-P000011

Observation. The abbreviation before “a boundary” in the lemma resembles Неπрс.

Candidates: Неπрс.; Неπр.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (490, 165, 775, 270). The bound 3^k agrees with counting choices link by link, but does not expand the abbreviation before “a boundary”.

Fresh source comparison: The full abbreviation, including its final curved stroke and period, is visible at 5×/10×. Неπрс. remains the closer historical candidate, without an expansion into a complete word. The counting explanation is separate.

GELFAND-0002-U004

Observation. After k a word is cancelled, with “links” inserted nearby.



Original source details: GELFAND-P000011

Candidates: точк. [points]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (1180, 165, 1425, 295). k links is natural for a contour; this only agrees with the clearly inserted word, while the cancelled candidate точк. remains separate.

Fresh source comparison: The inserted звеньев is readable. The cancelled line above it is crossed by dense strokes even at 10×; точк. is retained as a candidate rather than a recovered complete word.

GELFAND-0002-U005



Observation. Beneath the gain/loss line is a T-shaped junction of strokes.

Original source details: GELFAND-P000011

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (495, 320, 720, 425). The T-shaped mark lies by the gain/loss discussion; side labels or probabilities have not been invented from the neighbouring exponentials.

Fresh source comparison: The horizontal stroke, downward stem and a short separated right segment are visible at 10×. No arrowhead or independent label is established. The native sketch preserves the right-hand gap without assigning probabilities to its branches.

GELFAND-0002-U006



Original source details: GELFAND-P000011

Observation. A short cancelled word after “forbidden” is covered by heavy strokes.

Candidates: не [not]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (1270, 525, 1470, 610). The grammar of “returning is forbidden” does not determine the cancelled word; не is given only as a visual candidate.

Fresh source comparison: The beginning and a rising loop survive, but the middle is heavily inked over at 10×. The historical не candidate is neither declared certain nor expanded.

GELFAND-0002-U007



Original source details: GELFAND-P000011

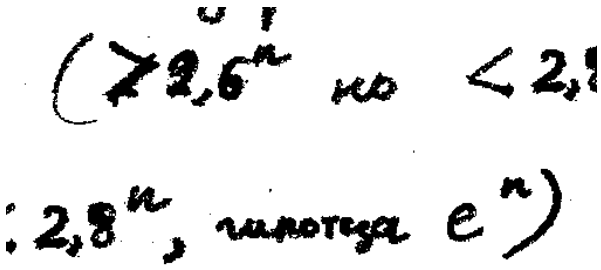
Observation. After “What is”, the first cancelled abbreviation resembles вер.; the next cancelled writing is heavily obscured.

Candidates: вер. ... [prob. ...]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (430, 590, 850, 680). The question about expectation makes вер. a possible cancelled abbreviation, but does not recover the next cancelled word.

Fresh source comparison: Two overlapping details preserve the two cancelled portions. The first admits вер.; the second remains unreadable beneath the cancellation at 10×. No probability question is reconstructed in place of the written expectation question.

GELFAND-0002-U008



Original source details: GELFAND-P000011

Observation. The first sign before $2{,}6^n$ has an additional lower stroke, allowing $\geq$ or >; next are $<2{,}8^n$ and “conjecture e^n”.

Candidates: $\geq 2{,}6^n$; $> 2{,}6^n$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (615, 835, 1545, 960). The exponential bounds resemble path counting, whereas the neighbouring question concerns distance; neither the quantity being bounded nor the initial sign has been corrected to remove this mismatch.

Fresh source comparison: The new detail confirms a heavy lower extension of the initial relation, but does not separate an equality bar uniquely from the pen stroke. Both relations remain. Separate finite walk calculations distinguish path counts from displacement without deciding the source claim.

GELFAND-0002-U009



Observation. The last abbreviation after “K is a nonlinear operator” has an initial H and an ending resembling мыц.

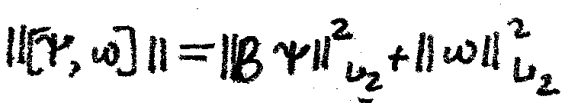
Original source details: GELFAND-P000011

Candidates: Немыц. [Nemyts.]; Намыц. [Namyts.]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000011.png; pixels (1320, 2240, 1590, 2365). The context K(u) for a nonlinear equation permits an operator defined pointwise by a nonlinear function, but does not determine the abbreviated name from expected letters alone.

Fresh source comparison: The initial letter and joined middle strokes remain visible at 10×. The vowel after H is not securely separated from its connection. Both abbreviated names remain; no full operator name is supplied.

GELFAND-0002-U010



Observation. The right side has squared norms; no separate exponent is visible after the closing norm on the left.

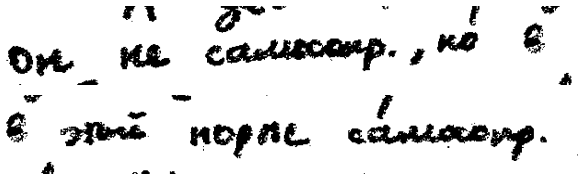
Original source details: GELFAND-P000012

Candidates: $\|[\varphi,\omega]\| = \|B\varphi\|^2 + \|\omega\|^2$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (360, 315, 1135, 445). The right side is quadratic when the pair (φ,ω) is scaled. This explains an expected square on the left, but it is not added without a visible exponent.

Fresh source comparison: The 5×/10× detail includes both closing norm bars and the blank exponent position before the equality. No separate 2 is visible there. The native right-hand squares are retained, and homogeneity is discussed only in the separate mathematical comparison.

GELFAND-0002-U011



Observation. The line reads “It is not self-adjoint, but in this norm it is self-adjoint.”

Original source details: GELFAND-P000012

Candidates: самосопр. [self-adjoint]; кососопр. [skew-adjoint]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (185, 480, 1390, 565). The matrix (0,I;−B^2,0) has a skew-adjoint structure in the energy inner product, agreeing with $A = -A^*$ in the preceding frame. The visible самосопр. has not been replaced by that deduction.

Fresh source comparison: Separate details show both occurrences with similar opening strokes, favouring самосопр. The matrix calculation does not turn those strokes into кососопр.; the historical alternative and source inconsistency remain visible.

GELFAND-0002-U012



Observation. Under M and “of t”, an underlined note at right resembles различные B.

Original source details: GELFAND-P000012

Candidates: различные B. [various B.]; разные B. [different B.]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (1600, 715, 1910, 820). The note is beside the solution set and the reference (**); its mathematical meaning does not distinguish the graphical readings разные/различные.

Fresh source comparison: The enlarged word has merged central strokes before the underlined B. The whole word and final B are retained in the crop; both historical word candidates remain.

GELFAND-0002-U013



Observation. In the parenthesis about a Banach manifold, writing between “Banach” and “space” is cancelled.

Original source details: GELFAND-P000012

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (295, 900, 585, 990). The

Banach manifold and smooth transitions do not allow the cancelled word, concealed by dense strokes, to be guessed.

Fresh source comparison: The 10× detail shows isolated gaps beneath two cancellation strokes, without a stable complete word. No missing adjective is inferred from the definition.

GELFAND-0002-U014



Original source details: GELFAND-P000012

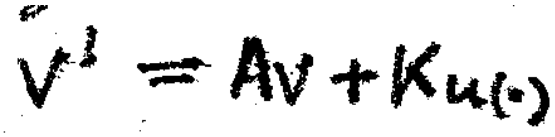
Observation. The side note begins “operator”, then мн./им. реш.; it ends with an arrow to H.

Candidates: мн. реш. → H [set of sol. → H]; им. реш. → H [им. sol. → H]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (1260, 1110, 1940, 1210). The domain of S_t is M and its values lie in H, so мн. реш. → H agrees with the visible scheme; им. remains an alternative because of the handwriting.

Fresh source comparison: The enlarged initial strokes admit мн., but their connections do not exclude the second historical reading. The full context shows the arrow and H; the short detail is used only for the initial abbreviation.

GELFAND-0002-U015



Original source details: GELFAND-P000012

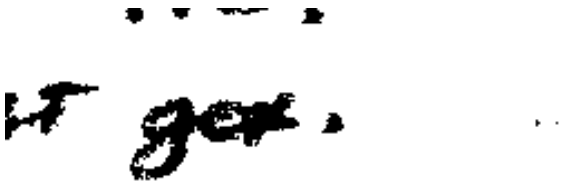
Observation. The formula is v′=Av+$K_{u(\cdot)}$; no separate v is visible after the last term.

Candidates: $v' = Av + K_{u(\cdot)}$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000012.png; pixels (690, 2020, 1160, 2130). Differentiating $K(u + \varepsilon v)$ with respect to ε normally gives the derivative applied to v. The absence of a separate v on the source line has not been corrected.

Fresh source comparison: The closing parenthesis is followed by blank paper at 10×. A separate factor v is not visible. The notation $K_{u(\cdot)}$ may itself be abbreviated; the conditional linearization formula in the commentary does not amend it.

GELFAND-0002-U016



Original source details: GELFAND-P000013

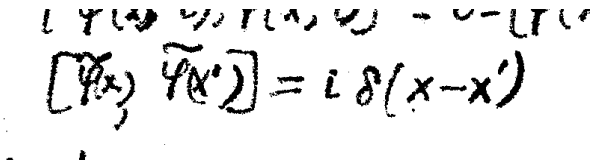
Observation. At the end of the second line of item 3 is an unclear abbreviation resembling зд.

Candidates: зд.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000013.png; pixels (680, 35, 945, 125). The list concerns vector fields and operators, but does not determine the meaning or expansion of the short зд. abbreviation.

Fresh source comparison: The descending initial loop and final thick stroke remain joined at 10×. The historical зд. candidate is retained without expanding it to “defined” from the sentence.

GELFAND-0002-U017



Observation. In the third commutator line no secure dot is separated above the second φ; an additional stroke follows δ, and the x/x′ arguments by φ are indistinct.

Original source details: GELFAND-P000013

Candidates: $[\widetilde{\varphi}(x), \widetilde{\varphi}(x')] = i\delta'(x - x')$

$[\widetilde{\varphi}(x), \dot{\widetilde{\varphi}}(x')] = i\delta(x - x')$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000013.png; pixels (625, 975, 1360, 1090). The preceding line already gives a zero equal-time commutator of two fields. The standard mixed relation would require a dot, but would not establish a dot or exclude a stroke after δ on the image.

Fresh source comparison: The corrected crop includes the whole tilde area above both fields. No isolated dot is securely visible above the second field. The stroke after δ continues downward like an opening parenthesis, favouring an unprimed δ, but the historical alternatives remain. The possible combination of two undotted fields and unprimed δ is not silently replaced by the standard mixed relation.

GELFAND-0002-U018



Observation. In parentheses beneath the definition of the Borel field is a short entry before “containing closed and open”.

Original source details: GELFAND-P000014

Candidates: min

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (120, 610, 435, 690). Minimality agrees with generation by open or closed sets, but min is retained graphically rather than expanded.

Fresh source comparison: The 5×/10× details show the connected min-like strokes. The candidate is retained as an abbreviation, with the neighbouring words in the full context.

GELFAND-0002-U019



Original source details: GELFAND-P000014

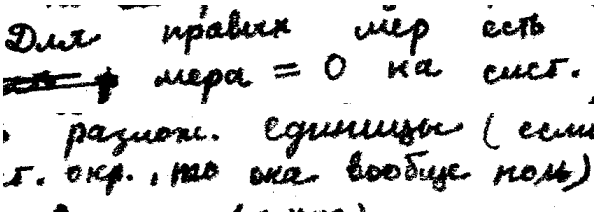
Observation. After “But” and before “Radon measures are more convenient”, short words are cancelled.

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (350, 915, 690, 990). The advantages of Radon measures do not determine the cancelled words; they have not been restored from the meaning.

Fresh source comparison: Both cancelled groups are visible at 10×, but dense horizontal and diagonal strokes prevent a full reading. Their number is descriptive rather than an inferred phrase.

GELFAND-0002-U020



Original source details: GELFAND-P000014

Observation. After “For right measures there is”, the abbreviation before “unity” is closer to разлож., though различ. is not wholly excluded; next comes “a measure =0 on a system of neighbourhoods, then it is zero everywhere”.

Candidates: For “right” measures there is разлож. единицы [decomp. of unity]; For “right” measures there is различ. единицы [различ. of unity]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (205, 960, 1815, 1100). A partition of unity could explain the passage from local vanishing to global vanishing. That supports разлож. as a candidate rather than an established reading.

Fresh source comparison: The full 3× context preserves the whole phrase. The short 5×/10× detail favours разлож. in its middle strokes but clips the final part of the word; it is not independent evidence for that ending. The description правых and both historical alternatives remain.

GELFAND-0002-U021



Original source details: GELFAND-P000014

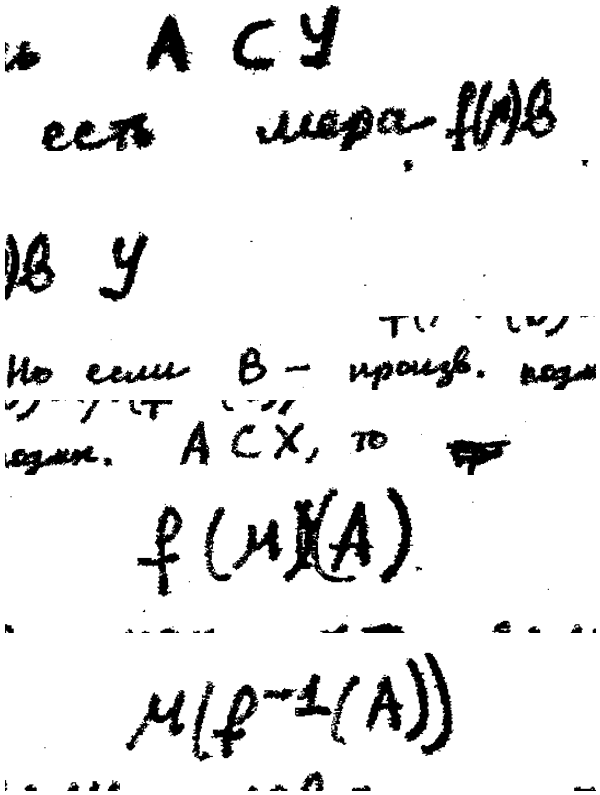
Observation. A short word before “Select” is cancelled.

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (765, 1190, 910, 1250). The following “Select on X and Y” does not determine the cancelled beginning.

Fresh source comparison: The dense central connection and descending loop remain, but no complete word is established at 10×. The cancelled entry is retained without a conjectural expansion.

GELFAND-0002-U022



Observation. The image-measure definition first has $A \subset Y$, then a formula with B, then a line with B and $A \subset X$, followed by formulas with A.

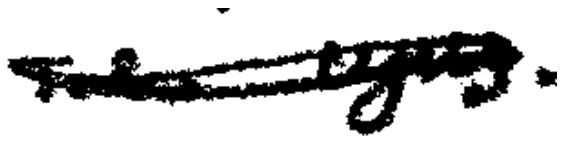
Original source details: GELFAND-P000014

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (470, 1320, 1380, 1735). For an image measure, sets in Y and their inverse images in X are natural. Nevertheless, the visible A/B and $A \subset X$ are retained without renaming.

Fresh source comparison: Separate enlarged details show $A \subset Y$ and $A \subset X$ with different last letters. Overlapping native crops retain the B formula and following A formulas. The domains and names are not reconciled by pushforward-measure reasoning.

GELFAND-0002-U023



Observation. Below the words that the converse is false, the start of a line is cancelled.

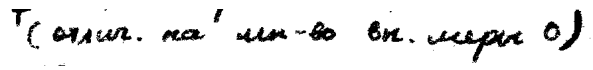
Original source details: GELFAND-P000014

Candidates: None proposed.

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000014.png; pixels (225, 1850, 550, 1940). The neighbouring measurability implication does not determine the crossed-out phrase; cancelled letters have not been invented.

Fresh source comparison: At 10× a few loops survive between the long cancellation strokes, but they do not form a stable complete phrase. The nearby implication is transcribed independently.

GELFAND-0002-U024



Original source details: GELFAND-P000015

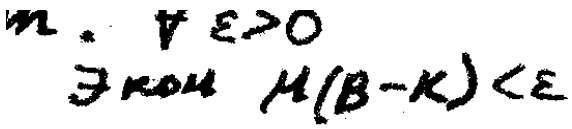
Observation. The parenthetical note after “The measure is the same” resembles отлич. на мн-во вн. меры 0.

Candidates: отлич. на мн-во вн. меры 0 [differs on a set of вн. measure 0]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000015.png; pixels (990, 245, 1880, 330). Zero outer measure can explain independence of the definition through E, but does not replace the uncertain source parenthesis by a rigorous formulation.

Fresh source comparison: The beginning отлич. and the ending вн. меры 0 are visible in overlapping details. The abbreviation вн. remains unexpanded in the diplomatic entry; the conditional independence argument is separate.

GELFAND-0002-U025



Original source details: GELFAND-P000015

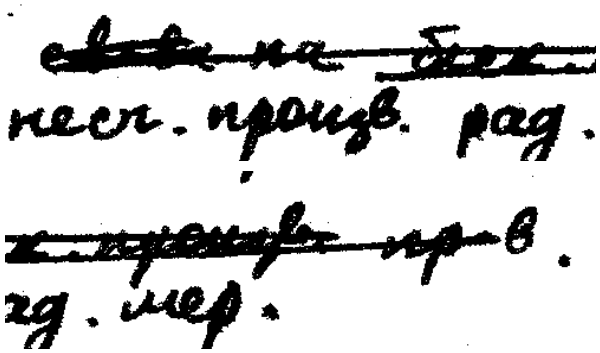
Observation. The lower condition reads μ Rad $\Longleftrightarrow$ ∀B compact ∀ε>0 ∃compact, $\mu(B-k)<\varepsilon$; no separate set letter or inclusion is visible after “compact”. The second letter in the difference is lowercase; the first letter’s case is uncertain.

Candidates: $\mu(B-k)<\varepsilon$; $\mu(b-k)<\varepsilon$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000015.png; pixels (1370, 1975, 1940, 2100). Inner compact regularity normally relates a compact set to a measurable set. The missing letter or inclusion and the visible B−k have not been supplemented.

Fresh source comparison: The new detail preserves the full difference and strict inequality. The second letter has the shape of k; the first favours B but its case remains marked. No missing compact-set name or inclusion is imported from the upper condition.

GELFAND-0002-U026



Observation. In the cancelled phrase beneath the horizontal stroke, an initial св-ва and a later бор. can be made out; the ending пр-в. is visible at right.

Original source details: GELFAND-P000015

Candidates: св-ва ... бор. ... пр-в. [properties ... Borel ... spaces]

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000015.png; pixels (900, 2180, 1820, 2330). The qualification about the support of an arbitrary Radon measure does not determine the densely cancelled phrase about spaces; visible fragments are only candidates.

Fresh source comparison: The two enlarged details show the initial and middle cancelled portions. Their long crossing strokes obscure complete words; the wider context retains the final пр-в. No complete topological qualification is reconstructed.

GELFAND-0002-U044



Observation. In $dx(E)$, both letters are on the baseline; the second appears lowercase, though its case is uncertain. The same notation is repeated after “its image”.

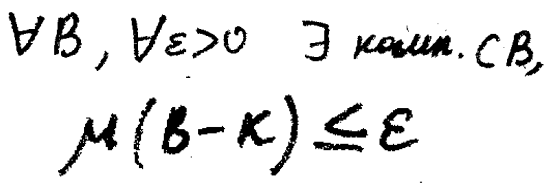
Original source details: GELFAND-P000015

Candidates: $dx(E)$; $dX(E)$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000015.png; pixels (1060, 115, 1290, 225). The notation for a measure on X and the Lebesgue-measure explanation clarify the sign’s role, but do not justify a subscript: dx/dX remain on the baseline.

Fresh source comparison: The 5×/10× detail clearly places x/X beside d on the baseline. The letter’s strokes are larger than some neighbouring lowercase writing, so capitalization is not settled. No subscript is inserted.

GELFAND-0002-U045



Original source details: GELFAND-P000015

Observation. The upper condition has ∀B ∀ε>0 ∃compact ⊂B; $\mu(b-k) \le \varepsilon$ visibly uses lowercase letters. No separate set letter appears between “compact” and the inclusion.

Candidates: $\mu(b-k) \le \varepsilon$

Historical note: The native crop at 1× and an unsmoothed 2× enlargement were re-inspected. Source chapter_001/01_part_000015.png; pixels (1120, 855, 1940, 1040). Inner regularity explains the difference’s mathematical role, but the lowercase b−k and absent compact-set name have not been reconciled editorially with ∀B.

Fresh source comparison: Separate details retain the gap after комп. before ⊂B, and the whole lower difference with its additional equality bar. The absent set name, letter cases and non-strict relation remain as written.

Editorial mathematical commentary

These comparisons do not amend the diplomatic text.

For a field k, consider $xv = yu$ in $\mathbb{A}^2_k \times \mathbb{P}^1_k$, with coordinates $[u : v]$. On $u = 1$, the equation is $y = xv$, with an affine v-line over the origin. On $v = 1$, it is $x = yu$, with an affine u-line over the origin. On their overlap, $uv = 1$, so the fibers glue to $\mathbb{P}^1_k$. Away from the origin, $[u : v] = [x : y]$ is unique. This model does not decide the source abbreviation Выбр., the ideal letter, or the general relation between symmetric and Rees algebras.

For distinct occupied sites, the symmetric sum over $i \neq j$ counts each unordered neighboring pair twice. Thus the written U is minus that pair count; the caption labels the positive count with lowercase $u(\theta)$. The case is retained. Under an explicitly chosen unordered-configuration convention, four sites in a chain with two particles give $Z = 3e^{1/T} + 3$; a 2×2 square with two particles gives $Z = 4e^{1/T} + 2$. In each case the weights divided by Z sum to one. Ordering the configurations multiplies Z by $N!$. The written $v = V_\ell / N_\ell$ is volume per particle. These finite checks do not prove the claimed thermodynamic limit.

Pressure sign and boundary term (source frames 4–5)

Assume explicitly that $v = V/N$, $f = N^{-1} \log Z$, and $F = -T \log Z = -TNf(V/N, T)$. At fixed N, T, the chain rule gives

$$p = -\frac{\partial F}{\partial V} = T \frac{\partial f}{\partial v}.$$

Thus a plus can be consistent with this convention. This does not establish the source's definition of f, the first argument's letter, or the stated monotonicity and convexity.

For a separate, explicit lattice model, let an interior subcube contain h^ν sites, with unit attractive nearest-neighbor bonds. Each of its 2ν faces has $h^{\nu-1}$ outgoing bonds. The coupling energy γ is minus the number of occupied crossing bonds, so

$$|\gamma| \leq 2\nu h^{\nu-1}.$$

With fixed $v = V/N$, $V = L^\nu$, and fixed h/L, this is $O(N^{(\nu-1)/\nu})$; in dimension three it is $O(N^{2/3})$. This explains a possible relation between the two powers without changing the tightly written exponent on V.

At fixed k, the configuration set splits into interior and exterior choices. At fixed positive T, a uniform bound $|\gamma| \leq CN^a$ bounds the coupled/decoupled partition-sum ratio between $e^{-CN^a/T}$ and $e^{CN^a/T}$. Taking its logarithm writes the ratio as $e^{\eta N^a}$, with bounded η. This coefficient is distinct from the configuration variable in this explanation. The source's θ/Θ remains uncertain.

Finite checks count 2, 2, 4, 8, 6 and 24 crossing edges for $(\nu, h) = (1,1), (1,3), (2,1), (2,2), (3,1), (3,2)$. All 126 four-particle configurations of a 3×3 box were also enumerated at $T = 1$, split according to occupation of the central site: 70 with $k = 0$, and 56 with $k = 1$. The partition ratios lie in the proved interval $[1, e^4]$. These checks do not prove the thermodynamic limit, phase-transition claims, or the ink readings.

Conventions, boundary counting and unresolved ink (source frames 6–10)

These calculations use stated assumptions. They do not replace uncertain manuscript signs.

Frame 6 writes $Z \approx e^{-Nf}$, whereas frames 3 and 8 use a positive limit $N^{-1} \log Z \to f$. Those are different sign conventions unless the meaning of f changes. Both source forms remain in the diplomatic body. In particular, conclusions about convexity are not transferred between conventions without checking their definitions.

For a finite occupied set in $\mathbb{Z}^\nu$, let E count unordered occupied nearest-neighbor pairs and let P count faces from an occupied site to an unoccupied site, including faces outside the finite box. Each occupied site has 2ν possible neighbors. Every occupied pair is counted twice, giving

$$2E + P = 2\nu N.$$

At fixed N and positive T,

$$e^{E/T} = e^{\nu N/T} e^{-P/(2T)}.$$

The common factor cancels after normalization. Thus using $U = E$ in $e^{U/T}$, or $U = -E$ in $e^{-U/T}$, produces the same perimeter weight under these explicit conventions. This explains a possible relation between frames 6–8; it does not settle a prime on Γ. If δ_x denotes the number of missing neighbors at occupied site x, then $\sum_x (2\nu - \delta_x) = 2E$, not E. The overwritten energy line is left unchanged because the local notation and relation remain uncertain. All 512 subsets of a 3×3 square and all 256 subsets of a $2 \times 2 \times 2$ cube were enumerated, including outer faces; the identity and normalized weights at $T = 1$ agree.

For a Euclidean disk of perimeter one, $2\pi r = 1$ gives area $1/(4\pi)$. For a three-dimensional ball of surface area one, $4\pi r^2 = 1$ gives volume $1/(6\sqrt{\pi})$. This computes the ball constants only; it does not prove the isoperimetric theorem or identify the overwritten $C/c/6$ in frame 9.

Put $q = e^{-1/(2T)}$ and $a = \nu/(\nu - 1)$, with $T > 0$, $\nu > 1$. The two candidates for the sum in $Q(T)$ contain respectively

$$\sum_{k \geq 2\nu} 3q^k k^a, \qquad \sum_{k \geq 2\nu} (3q)^k k^a.$$

The term ratios tend to q and $3q$. The first converges for every positive T. The second converges exactly when $3q < 1$, equivalently $T < 1/(2 \log 3)$; at equality its terms grow like k^a, and above it they do not tend to zero. This difference explains why the scope of the handwritten exponent matters. Convergence does not select a diplomatic reading, and these calculations do not prove the phase-transition criterion.

For a finite partition of a region of volume $V = Km$ into K cells of equal volume m, write g_j for the volume of G in cell j, and $g = \sum_j g_j$. Then

$$\frac{1}{K} \sum_{j=1}^{K} \frac{g_j}{m} = \frac{g}{V}, \qquad \text{hence some } j \text{ has } \frac{g_j}{m} \leq \frac{g}{V}.$$

If $g/V \leq (1 - \delta)/v$, this bounds the local fraction with denominator m. A denominator V, as visibly written at the bottom of frame 10, yields a different, weaker statement. The source denominator and its missing measure notation are not silently changed. The averaging identity was also checked in all 625 four-cell examples with each cell of volume four and integer occupied volumes from zero to four.

Finally, under a separate algebraic assumption, let A be a commutative algebra of functions over a field k, containing the constant functions. Evaluation at a point is a homomorphism $\mathrm{ev}_x : A \to k$: it preserves addition, multiplication and one. Constant functions make it onto, so $A/\ker \mathrm{ev}_x \cong k$ and the kernel is maximal. If an ideal vanishes at the point, evaluation factors through its quotient. This is a conditional explanation of the construction, not a reason to replace the visible target Φ in frame 7 or claim that every maximal ideal corresponds to an ordinary solution.

Walks, linearization and image measures (source frames 11–15)

These conditional calculations do not correct the diplomatic transcription.

On the square lattice, after a fixed directed first edge, a path that forbids immediate reversal has at most three choices for each next edge. Thus there are at most 3^{k-1} such words of length k with that first edge fixed, and $4 \cdot 3^{k-1}$ if the initial direction is free. Closed boundaries are a subset of those words, subject to their anchoring conventions. This explains a possible factor 3^k; it does not identify the abbreviated lemma or its exact objects.

For the separate walk that chooses uniformly among the three directions other than immediate reversal, let D_j be its unit increments and $S_n = \sum_{j=1}^{n} D_j$. Conditional expectation gives $\mathbb{E}(D_{j+1} \mid D_j) = D_j/3$, hence $\mathbb{E}(D_i \cdot D_j) = 3^{-|i-j|}$. Therefore

$$\mathbb{E}|S_n|^2 = n + 2\sum_{h=1}^{n-1}(n-h)3^{-h} = 2n - \frac{3}{2} + \frac{3}{2}\,3^{-n}, \qquad \mathbb{E}|S_n| \le \sqrt{2n}.$$

Exact enumeration through eight steps confirms this identity. The corresponding counts of self-avoiding paths through steps one to eight are $4, 12, 36, 100, 284, 780, 2172, 5916$, whereas the nonbacktracking counts are $4 \cdot 3^{n-1}$. Avoiding every previously visited site is a different rule from forbidding immediate reversal. These finite counts prove no asymptotic bounds 2.6^n or 2.8^n, and do not decide what the source means by returning or which quantity is bounded.

For a finite positive diagonal matrix B, put $H = \operatorname{diag}(B^2, I)$ and $A = \left(\begin{smallmatrix} 0 & I \\ -B^2 & 0 \end{smallmatrix}\right)$. Direct multiplication gives $A^*H + HA = 0$: the matrix is skew-adjoint in that energy inner product. The quadratic quantity $Q(\varphi, \omega) = \|B\varphi\|^2 + \|\omega\|^2$ satisfies $Q(cz) = |c|^2 Q(z)$, rather than the degree-one homogeneity of a norm. Neither observation adds a square on the left or replaces the written word “self-adjoint”. An unbounded operator would also require domain analysis, which this finite calculation does not provide.

If a differentiable solution family satisfies $u' = Au + K(u)$, differentiation in the family parameter gives $v' = Av + DK(u)[v]$. For example, $K(u) = u^3$ gives $DK(u)[v] = 3u^2 v$, since

$$\frac{(u+\varepsilon v)^3 - u^3}{\varepsilon} = 3u^2 v + 3\varepsilon u v^2 + \varepsilon^2 v^3.$$

The source’s abbreviated $K_{u(\cdot)}$ is retained without supplying a factor. Similarly, a zero commutator of two undotted fields cannot also equal a nonzero Dirac distribution, or its derivative, under the usual interpretation. This conditional consistency observation does not establish a dot, prime or argument in the handwritten line.

For a measurable map f and $\nu = f_*\mu$, a set in the completion of ν lies between measurable sets $E \subset A \subset F$ with $\nu(F \setminus E) = 0$. Pulling back these inclusions shows that $f^{-1}(A)$ belongs to the completion of μ, with the same completed measure. This proves the forward implication under explicit completion assumptions, without reconciling the source’s A/B or X/Y.

The stated example can be explained conditionally as follows. Assume $X \subset Y = [0,1]$ is non-Lebesgue-measurable and has outer Lebesgue measure one. Every Borel subset of $Y \setminus X$ then has measure zero. Consequently $\mu(X \cap E) = \lambda(E)$ is well defined on trace-Borel sets: two representatives differ by a Borel null set. For disjoint traces the representative overlaps are null; disjointizing them proves countable additivity. Under inclusion $i : X \to Y$, $i_*\mu$ equals λ on Borel sets. The particular set $A = Y \setminus X$ is not Lebesgue measurable, but $i^{-1}(A) = \varnothing$ is measurable. This gives the requested failure of the converse. “Non-Borel” alone would not imply “nonmeasurable”, so the source’s unrestricted choice of a non-Borel subset is not adopted as a theorem. The broad Radon and compactification claims remain outside these elementary checks.